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et de Technologie

Graduation Project report

To obtain the

National Diploma of Engineering
In
Applied Sciences and Technology

Subject:

Material Flow – Production site

Host organization:

VALEO Klimasysteme GmbH – Bad Rodach, Germany

Defended on : 01/10/2018

Realized by: **Iskander KHELIL**
Major: Instrumentation and Industrial Maintenance

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Academic Year : 2017/2018

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INSAT, Instrumentation and Industrial Maintenance

DEPOSIT AUTHORIZATIONS

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Academic Year : 2017/2018

Dedication

I dedicate this humble work, and all my achievements

To those who inspired it and will not read it

To the bright memory of my grandmother "Hafsa", who's been a rock of stability in my life and my greatest inspiration.
To the memory of my uncle "Mongi", my first teacher, who guided me and supported me all the way,
To the memory of my aunt "Badiaa", whose loving spirit sustains me still.
To the memory of my cousin "Karim", whose spirit will remain among us forever.

To the ones I look forward to,

To my father "Mourad", my role model, who taught me the real meaning of sacrifice, and stood by my side in each decision I made.
To my mother "Mouna", for nursing me with affection and love, and whose encouragements brought me success and honor.
To my dearest "Zeineb", my uniquely sweet, uniquely annoying and more importantly my uniquely lovable sister.
To my grandmother "Dalila", a wonderful combination of warmth and kindness.
To my aunts, uncles and cousins, for not giving up on me and giving me the best support I needed.
To my loyal friends, the kind souls I met and those who loved and believed in me, even at my lowest points.
To my country, for teaching me how to appreciate the little things, the hard way,
And foremost to Myself, I took the long path, but here I am finally at this juncture of my life..

"Traveler, there is no path,

The path forms itself as you walk it"

Acknowledgement

First of all, I would like to thank those who supported me with their encouragements, their unconditional help and for not giving up on me in the difficult times.

I wish to express my sincere thanks to my supervisor, Mrs Zahra BEN ACHOUR, for the time afforded, for her advices and for her guidance.

I would like to express my deepest gratitude to my supervisor at Valeo, Mr. Thomas FRITZ, for making sure I was integrated and well involved in the project despite the language barriers.

I also want to thank my colleagues who made my integration smoother than expected, and for doing their best to make me feel home.

I place on record my sincere gratitude to the "Family" I lived and enjoyed my days with in Germany, I am extremely indepted to them, for everything they have done.

Finally, I take this opportunity to thank the honorable members of the jury for agreeing to review this work, hoping that this report reflects the quality of the work carried out during this graduation project.

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List of Acronyms :

VPS: Valeo Production System.

DH: Distribution Housing.

AI: Air Intake.

AGV: Autonomous Guided Vehicle.

HVAC: Heating, Ventilation and Air Conditioning.

MRA: Modular Rear Architecture.

2-Z: 2 Zones.

4-Z: 4 Zones.

LHD: Left Hand Driver.

RHD: Right Hand Driver.

BR: Bad Rodach site.

WIP: Work In Process.

Assy: Assembly.

TAHC: Thin Aluminum Heating Core.

EOLT: End Of Line Tester.

EBTT: Electronic Built To Truck.

SAP: Systems, Applications and Products.

General introduction:

Given the increased competitiveness between industries in the different fields, and the pursuit to insure a higher productivity in the shortest possible time and with minimum cost, every member of the field is trying to adopt solutions and tools to satisfy the customer demand, and also to anticipate the unexpected changes in the market by offering more flexibility.

Therefore, the automotive field in particular, follows the same pace while facing the increase of the customer orders, and the response to this demand that must be agile and effective.

Valeo, a world leader in the automotive equipment sector, stands out to face the high level of orders and customer demand by being able to adapt to the market and adopt the production capacity that meets the expectations of its customers.

In this perspective, my internship took place in Valeo Climate Control Systems, in Bad Rodach Germany, within the assembly production department APU-M: Autonomous Production Unit– Montage, during the period between May 1st and October 31st. And my graduation project, entitled « Material Flow in the production Site », has as main objective the adaptation of the new cooling systems assembly lines for the new « Daimler » business, by optimizing the internal material flow as the customer demand increases.

The project, called MRA2, is considered as one of the biggest projects to be implemented in the plant, and my task covers the following topics:

- Layout and adjustment of the new assembly lines.
- Material and information flow studies.
- Application of the correct tools for those issues.

To adopt the best approach to the project, my work was divided between the observation and studies, analysis, conception, flow analysis and work in progress calculation.

Thus, this report is composed of 3 main chapters that describe my work from May 1st until August 31st. The first chapter is devoted to the presentation of the host company « Valeo and the project scope. The second chapter will describe the final product and a global presentation of the project as a whole and will treat the tools of lean manufacturing and management valuable for the project. And the third and last chapter is devoted to my studies, modifications and planning of the production to make it as optimized as possible, including the assembly of the parts, the material flow and the potential information flow studies development process.

Chapter I:

Presentation of the host company and project framework.

Introduction :

In this chapter, we will present at first the host company “Valeo” worldwide, and then a specific presentation of “Valeo - Climate Control Systems” plant in Bad Rodach, Germany.

The next point will be a demonstration of the project framework and the product’s description, including the steps and assembly lines positioning along the project’s cycle.

This chapter will be concluded with an enhancement of the project scope in order to set up the new assembly lines specifically for the new project by exposing the different specifications concerning my work.

I.1. Presentation of the host company: Valeo:

I.1.1. Generalities:

Valeo is an automotive supplier and strategic partner to all automakers worldwide, focusing on the design, manufacture and sale of equipments, integrated systems and modules for the automotive and heavy weights vehicules.

The group was founded in 1923 in Saint Ouen (France), and the expression “Valeo” is a latin expression which means “I’m fine”.

Since the beginning of the 1980s, Valeo continues to grow, and today, Valeo is among the world’s automotive and commercial vehicle suppliers leaders.

Valeo’s main mission is to offer its automaker customers innovations enabling automobile use to evolve. The group’s ambition is to be a leader in all its business activities, and therefore the indisputable technological benchmark among automakers. [1]



Figure 1. Valeo's main customers.

I.1.2. Valeo Worldwide:

Currently, Valeo is operating in 33 countries, as shown in the repartition map in figure 2, and has more than 111,860 employees devided between 184 production sites including 20 research centers, 35 development centers and 15 distribution platforms with generated sales of 18,6 billion dollars in 2017 and expected to exceed 27 billion dollars by 2021. [2]



Figure 2. Divisions of Valeo group: Business activities structure.

I.1.3. Divisions of the Valeo group: Business activities structure:

Valeo is made up of 4 main business groups and aftermarket activities :

- ***The comfort and driving assistance systems business group:***

(3,6 Billion € of sales (19% of the group sales) / 27 factories):

This business group develops interface systems between the driver, the vehicle and the environment, which help to improve comfort and safety. 3 product groups are produced including interior control, connected car and driving assistance.

- ***The powertrain systems business group:***

(4,3 Billion € of sales (23% of the group sales) / 55 factories):

This business group develops innovative powertrain solutions aimed at reducing fuel consumption and CO2 emission. It produces electrical systems, transmissions systems, combustion engine systems and electronics.

- ***The Thermal Systems Business Group:***

(5 Billion € of sales (27% of the group sales) / 62 factories / 24.200 employees):

This business group develops and manufactures systems, modules and components to manage the thermal energy of the powertrain and provide comfort inside the cabin for each passenger.

Thermal systems has four product groups: Climate control, Powertrain thermal systems, Climate control compressors, Front-end modules.

- ***The Visibility Systems Business Group*:**

(5,8 Billion € of sales (31% of the group sales)):

This business group designs and produces innovative systems which offers the driver perfect visibility, thereby improving the safety of both driver and passengers. It produces mainly lighting systems and wiper systems. [3]

- Comfort and driving assistance systems.
- Powertrain systems.
- Thermal Systems.
- Visibility Systems.

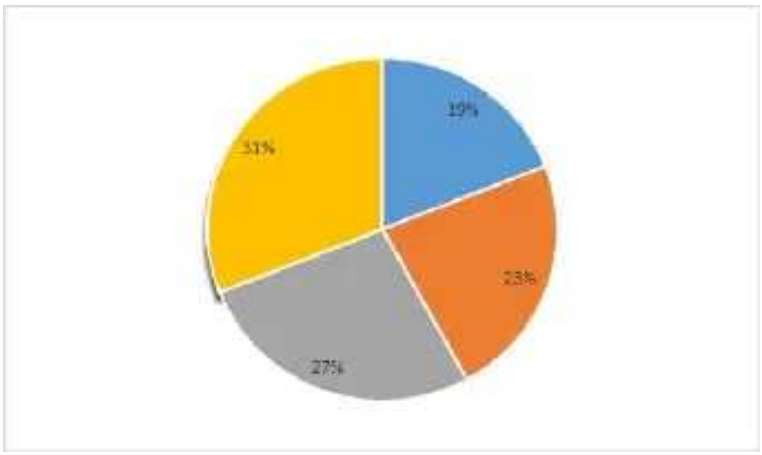


Figure 3. Valeo Business groups.

I.1.4. Valeo's strategy:

Valeo's strategy is built on two priorities: innovation with reduced CO2 emissions and improved vehicle performance, as well as intuitive driving and development in high growth regions (especially Asia). Valeo is both open (open innovation) and rigorous. It possesses the capacity to innovate (technological company), but also to ensure reliable and high quality large scale production (industrial company).

In order to satisfy the customers, by fulfilling their needs in term of quality, cost and delivery, the group applies a 5 axes methodology, represented in figure 4, that is rigorously implemented by all of its employees. It includes the following parts :

- **Involvement of personnel:**

The involvement of staff is not only about recognizing their skills but also enhancing them through training and encouraging them to make suggestions for improvement and participating actively in the functioning of their autonomous team.

- **The Valeo Production System (VPS):**

Improving the productivity and the product quality is the objective to attend in the quest of the customer satisfaction. For this reason, Valeo has developed its own production system, based

on pull-flow organization, flexible production resources, the elimination of all non-productive operations, and stopping production at first non quality incident.

- **Constant innovation:**

To develop its systems and products, Valeo has set up an organization based on project teams and simultaneous engineering of products and processes. This allows to design innovative and high quality products, easy to manufacture and cost effective with reduced development times.

- **Supplier integration:**

Valeo establishes and maintains long-term close and friendly relations with a limited number of suppliers, chosen from among the best in the world, with whom it maintains mutually beneficial long-term relations which contributes to develop productivity plans and improves the quality of the work.

- **Total quality:**

In order to meet customer expectations, in terms of products and service quality, total quality is required throughout the group and from its suppliers.



Figure 4. 5 Axes methodology: client satisfaction strategy.

I.1.5. Valeo in Germany:

Valeo has been in Germany for many decades and German customers account for more than 30% of the group sales worldwide. Around 6500 employees are divided between the 20 different Valeo sites in Germany (assembly, production, research and development). [4]

I.1.6. Bad Rodach Site: Climate Control Technology:

The site in Bad Rodach is part of the Thermal Systems (THS) Business Group and is located in Franconia, close to the border with the state of Thuringia.

This Valeo site is a production site and an important development center for thermal systems (Region Thermal Systems North Europe). It also features modern noise test benches, climate control test benches, test benches for performance tests of all kinds, 3D camera scans for parts measurement and design development for operating elements.



Figure 5. Valeo Climate Control: Bad Rodach site.

As a part of Valeo Climate Control Technology, this plant specializes in the research, development and production of high complex HVAC (Heating, Ventilation and Air Conditioning) systems, and is considered as a heating and air conditioning leading provider where thermal systems, such as complete climate control systems as well as individual air conditioning components, are being produced and designed for national and international automobile manufacturers.

I.2. Project's aim and steps:

I.2.1. Project's aim:

After a global study of the project, the assembly process and the vision of the company, our aim is to create an optimal work area for the future project with a "futuristic" vision in the assembly area, that will enable the implementation of the new assembly lines, in the space dedicated, while making sure of the synchronisation between the physical material flow and information flow. The appendix 1.Valeo Production departments shows the different departments of the plant through which the products are created, stored and assembled.

The targets to reach during my internship are:

- Defining a layout in the assembly area.
- Optimizing the layout and adapting it to the space allocated by using lean tools.
- Implementing an efficient physical material flow for the assembly area.
- Projection of an information flow synchronized with the physical material flow in the assembly area.

I.2.2. Project scope and steps:

The project will be focusing mainly on 3 big parts:

- **Layout adaptation:** assembly lines implementation.
- **Material flow studies:** planification and volume of production.
- **Information flow:** development and synchronization of production.

Based on this, the steps of my work are as following:

1. The measurement and understanding of the general process and layout including the machines and steps of assembly in the assembly area.
2. A study of the process flow of the existing and the future projects.
3. Calculation of the Work In Process for the MRA2 project.
4. Studies and elaboration of a smooth material flow.
5. Creating a planning of the material flow in the assembly for MRA2 project.
6. Development of a suitable and optimized information flow.
7. Concept development of automatic assembly and transport.

While going through the project, we should keep in mind as main objective the reduction of the costs for an economic production and higher production capabilities of HVAC units, with the development of a concept for automated production lines.

Conclusion :

In this chapter, we presented from one side, the host company Valeo, and in particular the Bad Rodach site, and the production speciality of this plant, and from another side our project's framework and globally the different topics to be treated during my internship.

The different points will be detailed in the next chapters.

Chapter II:

Product description and project presentation:

Introduction :

This chapter is divided into 3 parts. It will be devoted at first to describe in general our finally purshased product HVAC unit, its utility, the functionalities and the different existing versions, followed by the specifications of the aimed MRA2 project and the means of assembly of the different parts of the HVAC, and thereby the particularity of our assembly lines and the correspondent varieties. The specifications of each step were studied individually which will help us later understand the particularities of the process. The last part is dedicated to the lean tools used or will be used during the project.

II.1. Product description:

II.1.1. HVAC: Heating, Ventilation and Air Conditioning:

HVAC is the technology responsible for the automatic air regulation and comfort, by controlling the climate conditions in the vehicle.

Nowadays, HVACs have become a standard equipment in automotive market providing the thermal comfort inside the vehicle.



Figure 6. HVAC Unit for automotive use.

This system heats or cools, cleans, regulates or dehumidifies the air drawn from outside the vehicle and distributes it to the passengers at the required temperature allowing an acceptable air quality and a comfortable climate inside the cabin.

II.1.2. HVAC functionalities:

The operation of heating, ventilating or air conditioning is based on the circulation of cold or hot air : Outside air is picked up at the base of windshield and then blown inside the vehicle. The air is ventilated, filtered, cooled or heated, mixed and distributed into the cabin.

HVAC system has 3 main functions:

- **Heating:** This operation is carried out via a central heating system. The surrounding air enters the unit and is warmed by a heater core. Heat is then distributed through a system of air ducts to air distributor inside the vehicle.
- **Ventilation:** The ventilation function of the HVAC system maintains the best air quality inside the vehicle. This function is ensured by exchanging the air from outside after removing humidity or impurities (oder, smoke, heat, dust...) from outdoor air, and oxygen replenishment.
- **Air conditioning:** The operation is done by using a refrigeration system ensuring the removal of humidity and heat from the air to have cooled air in output. Cooled air is then distributed inside the cabin via a distribution system.

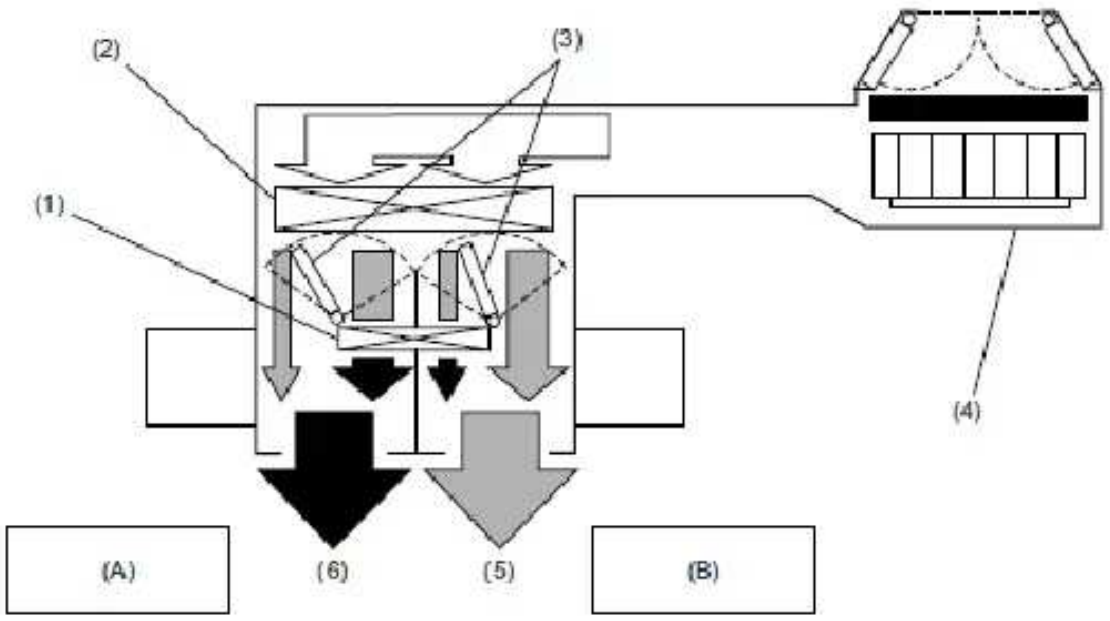


Figure 7. Air circulation in the HVAC unit.

(1) Heater core. (2) Evaporator.
(3) Air mix door. (4) Blower unit.
(5) Cool air. (6) Warm air.
(A) Driver's seat. (B) Passenger's seat.

II.1.3. HVAC varieties:

With the emergence of the new technologies and the desire for more comfort and luxury, and as cars are more and more automated, climate control systems are following the same trend.[5]

In fact, many varieties appeared to satisfy the customer needs and to be up to date with this evolution.

The differences during the production of HVAC systems depend from many criterias and specifications from a brand to another and within the same brand we can find different varieties such as the driver side mode (left or right), the type of vehicle (standard, hybrid or electrical) and the number of comfort zones (from 1 to 6 zones) in the vehicle, which offers different levels of comfort and more personalized cooling / heating control.

The specifications of HVAC systems are different from a brand to another, and within the same brand, we can find different varieties and comfort zones numbers. So we can say that the number of comfort zones dedicated for separate climate control inside the vehicle is no longer limited to large vehicles or luxury brands.

II.2. MRA2 Project:

II.2.1. Project description :

My internship took over around the MRA2 project, a new « Daimler » business, with the intention to produce HVAC units for the new vehicles categories : BR206 – C-Class Carline, BR214 – E-Class Carline and EVA2 Electrical Vehicle.



BR206
C-class carline

BR214
E-class carline

EVA2
Electrical vehicle

Figure 8. Cars categories for the new Daimler business.

This project will initially see the light starting from 2020 and lasts until 2030, following MRA1 project.

II.2.2. MRA meaning:

In 2014, Mercedes-Benz introduced a new modular automotive manufacturing concept known as Modular Rear Architecture (MRA). This modular concept is also available for the front-drive Automobile (Modular Front Architecture-MFA). The concept is so concerted that different car models with different wheelbases but the same components can be built on one platform. The first model made on the MRA platform was the C-Class. The platform was designed to make the final assembly of automobiles more flexible so that Mercedes can mount all rear-wheel drive vehicles on this platform.

II.2.3. Project varieties:

The product's varieties depend mainly on:

- The type of the vehicle (Standard / Hybrid or electrical).
- The number of comfort zones (2 zones and 4 zones).
- Driver mode (Left or Right Hand Driver).

The figure 9 is a representation of all the MRA2 HVAC varieties. ⇒**Total=12Varieties.**

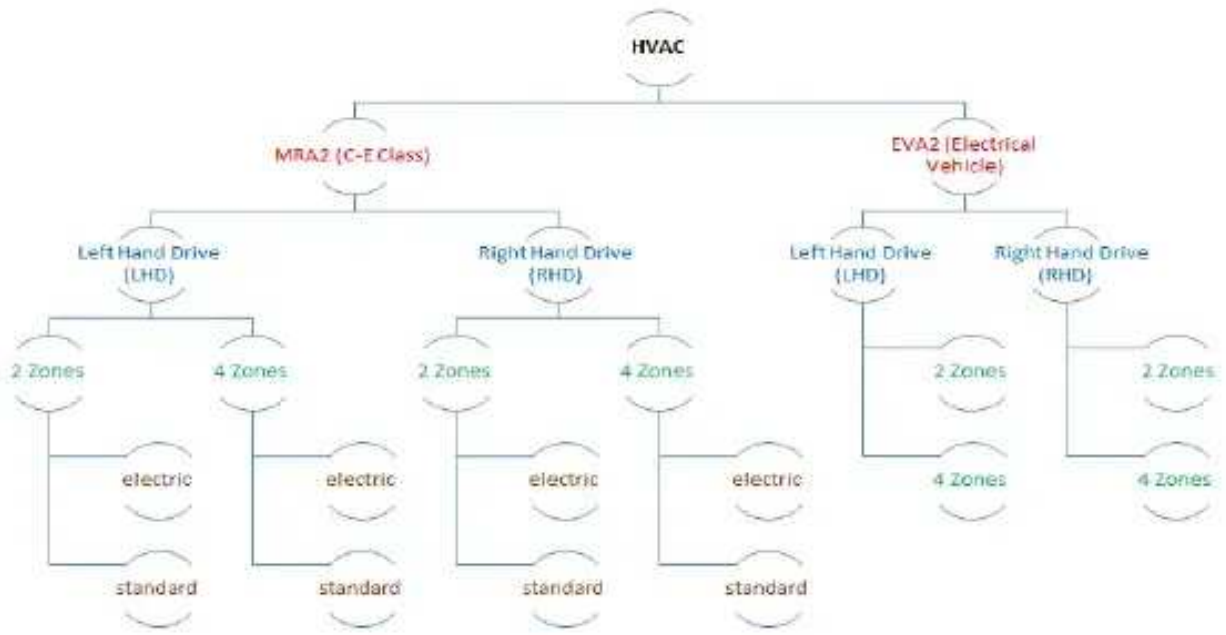


Figure 9. HVAC Varieties for MRA2.

II.2.4. HVAC parts:

As shown in the figure 10. The HVAC units are composed of 3 pre-assembled main parts:

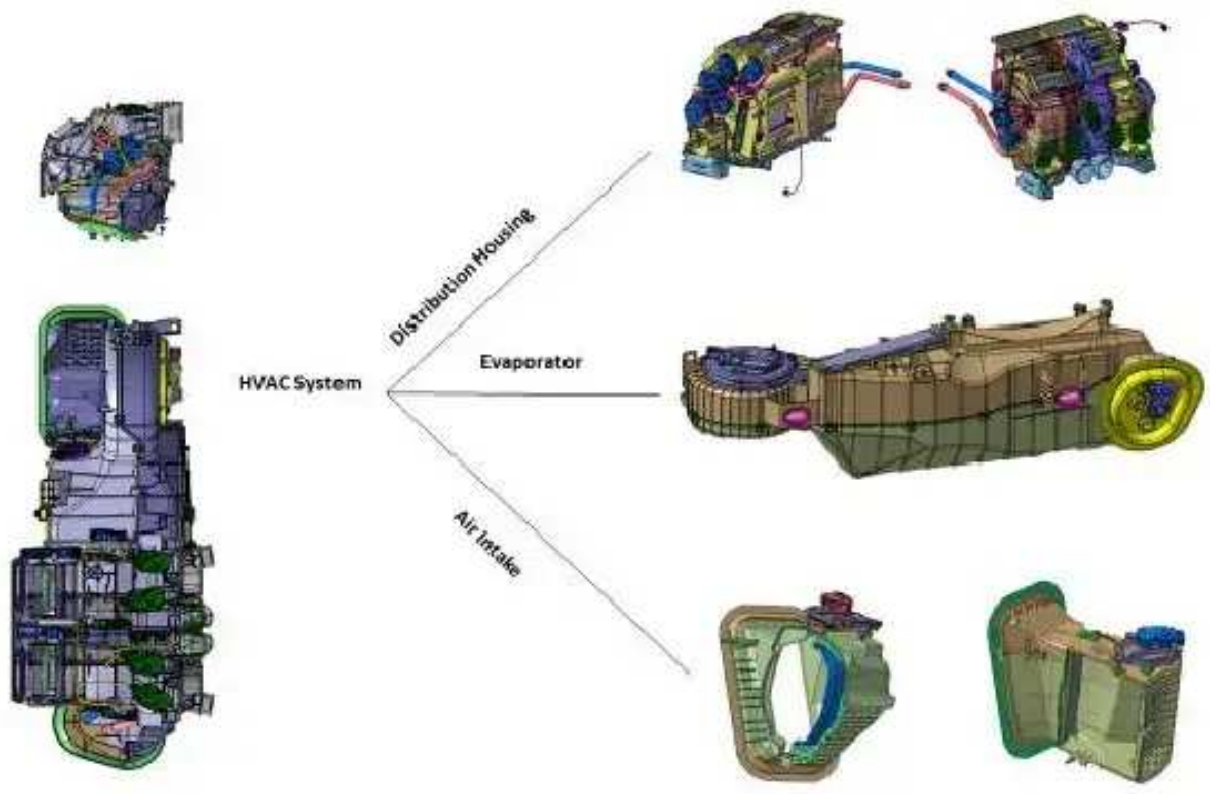


Figure 10. Main pre-assembled modules of HVAC.

II.2.4.1. Distribution Housing:

This part of HVAC is responsible for the air distribution inside the vehicle and the control of shutters via the actuators. The distribution housing is the part on which depends the number of zones for individual temperature adjustment.

We distinguish 8 different varieties of the distribution housing: 4 Varieties are produced locally (Bad Rodach site) and 4 are brought from another site (Rakovnik site in Czech Republic), as represented in the table 1 and the Appendix 16.

<table>
<tr><td rowspan="4">MRA2</td><td rowspan="2">2 Zones</td><td>LHD : Left Hand Drive.</td></tr>
<tr><td>RHD : Right Hand Drive.</td></tr>
<tr><td rowspan="2">4 Zones</td><td>LHD : Left Hand Drive.</td></tr>
<tr><td>RHD : Right Hand Drive.</td></tr>
<tr><td rowspan="4">EVA2</td><td rowspan="2">2 Zones</td><td>LHD : Left Hand Drive.</td></tr>
<tr><td>RHD : Right Hand Drive.</td></tr>
<tr><td rowspan="2">4 Zones</td><td>LHD : Left Hand Drive.</td></tr>
<tr><td>RHD : Right Hand Drive.</td></tr>
</table>

Table 1. Distribution Housing Varieties.

II.2.4.2. Evaporator:

The evaporator is the part responsible for filtering the air and removing the heat and maintaining thermal comfort even when the engine is switched off (electrical vehicle case).

This part represents the 6 different varieties produced in Bad Rodach represented in Table 2.

MRA2	LHD : Left Hand Drive.	Shut Off
		Standard
	RHD : Right Hand Drive.	Shut Off
		Standard
EVA2	LHD : Left Hand Drive.	
	RHD : Right Hand Drive.	

Table 2. Evaporator Varieties.

II.2.4.3 Air Intake:

Air intake pre-assembled part is the part responsible for bringing the air into the car's engine. It's containing some components insuring the air filter isolation, providing stable air temperature. It is composed of 4 versions as shown in table 3.

MRA2	LHD : Left Hand Drive.
	RHD : Right Hand Drive.
EVA2	LHD : Left Hand Drive.
	RHD : Right Hand Drive.

Table 3. Air Intake Varieties.

II.2.5. Assembly lines:

The assembly process of HVAC units is carried out by relating 3 main different pre-assembly lines (DH Assy, AI Assy, Evaporator Assy) to the final assembly line (Back Bone).

Globally, the space allocated for our project is estimated to cover a surface 1138 m², and the assembly lines are going to be implemented separately, at first, in the correspondent space. The annexe 2. Assembly Area : APU-M is a representation of the APU-M area (assembly area) and the space allocated for MRA2 project. The assembly lines are respectively:

II.2.5.1. The distribution housing assembly line:

This assembly line, represented in the figure 11, is highly automatic, and it will include the assembly of housings, flaps, levers… The steps of assembly are fulfilled in 17 workstations, shared between 2 modules related to each others.

- Section 1: from station 1 to 13: Module A:

Considered as the pre-assembly of the distribution housing, including the merging and greasing of main housing, flaps and levers.

- Section 2: from station 14 to 17: Module D:

It is considered as the final assembly of kinematics, heat exchangers, wire harness and other small parts, as well as mid-line testing of the sub-module.
The steps of assembly in each workstation are detailed in the appendix 3.
This module will be managed by 4 operators and the cycle time is expected not to exceed 18s.

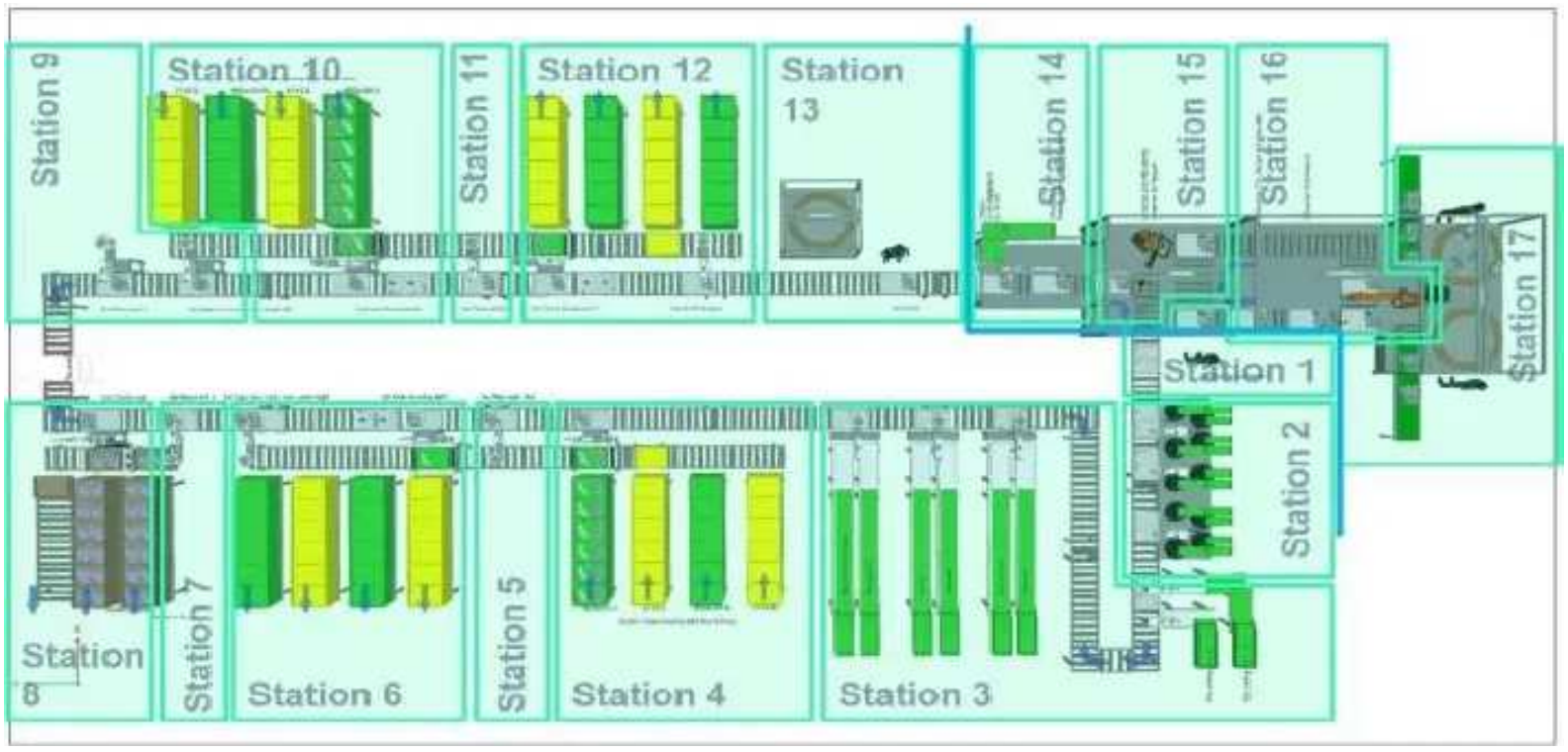


Figure 11. Distribution Housing assembly line.

II.2.5.2.The Air Intake assembly line:

The Air Intake Module (Module C), as shown in the figure 12, is composed of 2 sections:

- Section 1: From one hand, this assembly line is a high-automated solution for the part

version BR206/214 (MRA2), and from the other hand, it is a manual pre-assembly for the part version EVA2.

The main function of section 1 is: processing the assembly of flaps to the prepared housings including the levers.

- Section 2: A high automated solution capable to process both versions of Air Intake:

Synergy or fusion for the assembly of the kinematics and the actuators and also a combination of both versions and ends with an automatic screwing and packaging of the module to a box.

Each version goes through 7 steps to get a pre-assembled air intake semi finished part.

The steps of assembly in each workstation for both MRA2 and EVA2 are detailed in the appendix 4. This line is managed by 2 operators and the cycle time is estimated to be 18s.

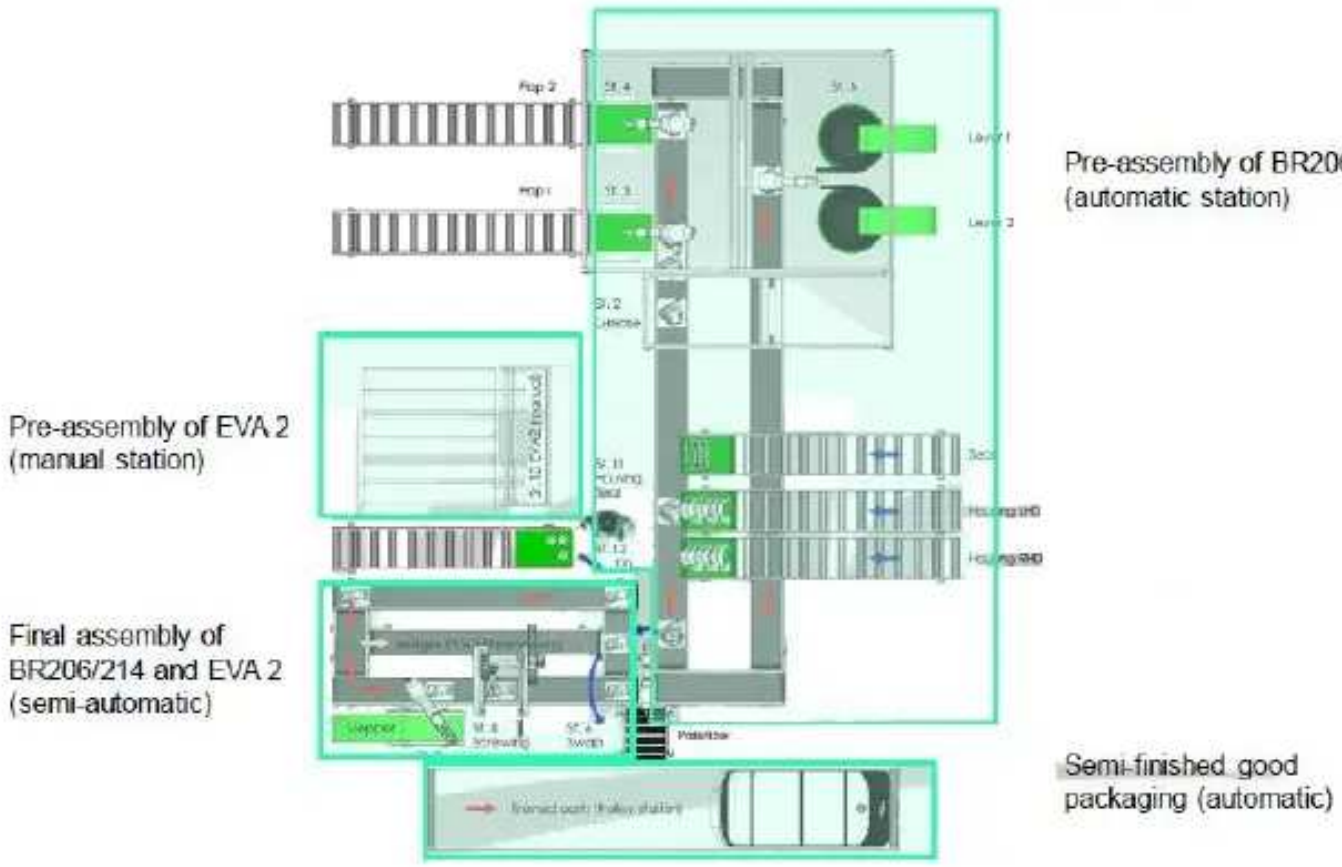


Figure 12. Air Intake assembly line.

II.2.5.3. The evaporator assembly line:

The assembly line is composed of 5 workstations for the different variants MRA2 (LHD/RHD) and EVA2 (LHD/RHD).

The details of assembly are represented in the appendix 5 and the figure 13 is a layout representation of the evaporator pre-assembly. The cycle time of this module is estimated by 38s and the line will be managed with 2 operators.

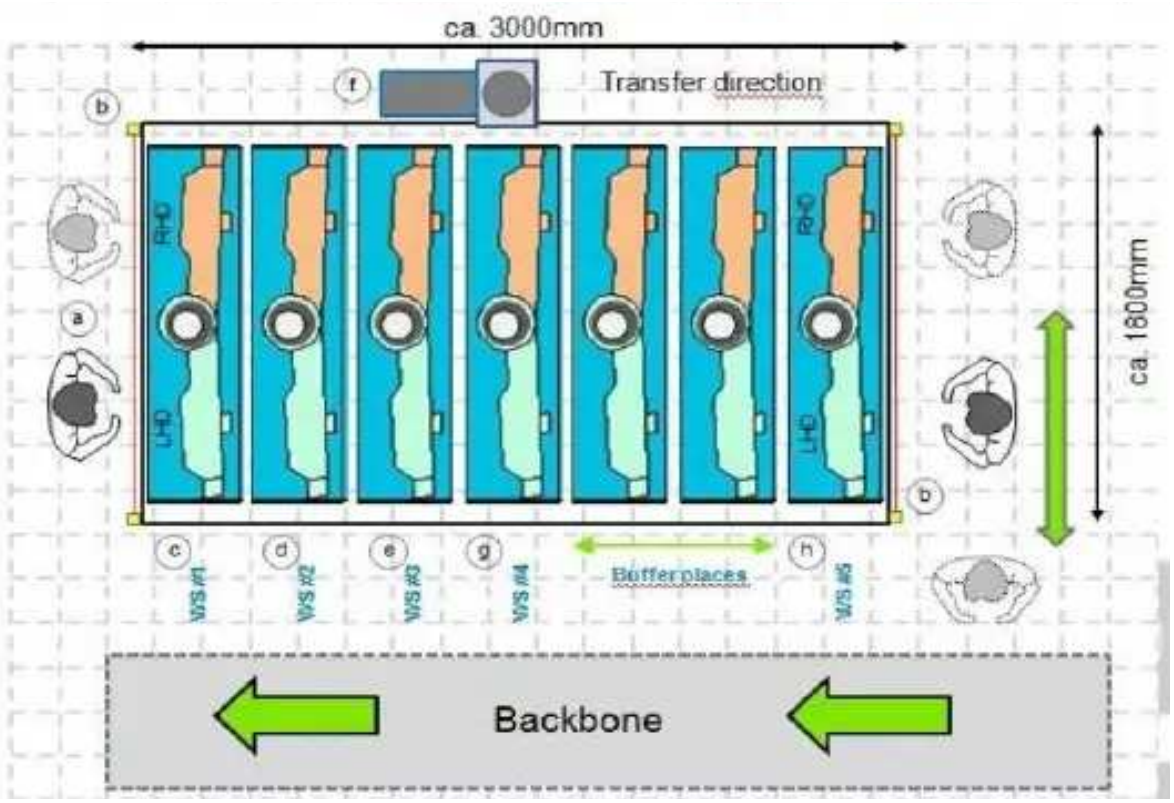


Figure 13. Evaporator assembly line.

II.2.5.4. The Back-Bone assembly line: (Module E):

The Back-Bone, represented in figure 14, represents the point of meeting of all 3 pre-assembly lines to have a final assembled HVAC unit in output.

The cycle time is estimated to be 38s and managed by 2 operators (shared with the evaporator module). The weight of the final assembled HVAC unit in output is varying from 6 Kg to 12 Kg depending on the variety. The details of assembly of the final HVAC are represented in the appendix 6.



Figure 14. Backbone assembly line.

II.3. Lean tools :

The operation of implementing new layouts is usually realized by relating the right tool to the correspondent operation that generally turns around people management, equipment and material management, and inventory management.

The tools used in our project will be mentioned in the same order followed while implementing the assembly lines, so the main pillars are:

- The Layout: implementation of the assembly lines and the buffer area.
- The Material flow: loading, storing and routing.
- The Information flow: scheduling and planning.

In the following, we will be dividing the lean tools according to the mentioned 3 pillars:

II.3.1. Layout pillar: work environment and housekeeping:

After studying the product specifications (HVAC) and how globally the assembly process should be realized, we need to prepare our work environment in terms of space and modules/facility planning.

II.3.1.1. 5 S: Housekeeping:

In order to comply with the strategic requirements of total quality and productivity improvement, Valeo has introduced its own production system VPS, boiling down to the application of 5 S as a first step.

The 5S is a Japanese management technique developed at Toyota, it is a standardized process that when properly implemented reduces waste and optimize productivity by creating and maintaining an organized, safe, clean and efficient workplace.

The pillars of the 5S method, as shown in figure 15, are based on the abbreviation of the following five Japanese verbs: Seiri, Seiton, Seiso, Seiketsu, Shitsuke.

- **Seiri / Sort:** Separating the essential from non essential items / operations and eliminating useless operations.
- **Seiton / Straighten:** organizing the essential materials in the right place where and when they are needed.
- **Seiso / Shine:** cleaning the work area.
- **Seiketsu / Standardize:** Making 5S a habit by establishing a system to maintain it.
- **Shisuke / Sustain:** Establishing a safe and sanitary work environment (safety).



Figure 15. 5S methodology : Housekeeping

Many benefits can be gained from the implementation of the 5S methods into a work area, it will help to control costs and eliminate the wastes, avoid unplanned downtime and reduce in-process inventory. [6]

II.3.1.2. Mudas/Muri/Mura:

As mentioned, 5S method principles are effective tools to reduce or eliminate wastes and improve quality or increase efficiency, so Mudas (wastes) are to be kept under control while implementing the lines.

In fact, the Mudas are non value adding activities, and according to Taiichi Ohno, the father of the Toyota Production System, the Mudas can be classified into seven types of wastes:

- **Over production:** Producing more or faster than needed.
- **Waiting:** keeping a worker idle by waiting for material or people.
- **Transportation:** moving materials or products excessively.
- **Over processing:** doing more than is required.
- **Inventory:** excess raw material, work in process or finished goods.
- **Defects:** Repairing errors.
- **Motion:** Acting or moving without adding value.

Mura: the waste of unevenness: the Mura result from the fail to smooth of demand or unbalance between the production of the assembly lines.

Muri: The waste of overburden: Muri is the cause of overburden, by this we mean to give unnecessary stress to the employees and the process. This is caused by Mura and other failures in the system.

To resume, Mura causes Muda, the seven wastes, which are symptoms of our failure, so, to tackle Mura and Muri is the way to eliminate the wastes. [7]

II.3.2. Material flow pillar:

Intern physical material flow analysis between assembly lines is a way to know how and where the material is moving inside the area of assembly.

For this reason, LEAN tools are going to be the way to understand the flow, to have a global view and to present other perspectives.

II.3.2.1. Value Stream map:

Value Stream Map (VSM) is a way to represent the flow of material through the organization as well as the flow of information, which enables to notice if there are difficulties at a certain point of our process (delays, excessive inventory, transport…).

By representing the current map, we can make improvements on the state of our flow, to end up with an ideal future Value Stream Map. [8]

II.3.2.2. Fish bone diagram:

It's a representation of the main problems or possible difficulties surrounding the project.

This tool shows the root of any problem that can be related to the production, and presents the possibilities to solve them.

Fish bone (Ishikawa) was used for two different purposes: at first to give a global view of the project and how the lines are related to the final assembly line, including the flow of material and the correspondent cycle/takt time, and then to represent the relationship between the problems (impact) and all possible factors (causes).

II.3.2.3. Process Flow diagram:

A process flow or a flow chart is a picture of the separate steps of a process in sequential order. It's a representation of sequence of actions, material entering or leaving the process (as inputs or outputs), time involved at each step and process measurements… [10]

This is a generic tool that can be adapted for a wide variety of purposes, in our case, this tools will give us a general overview of the different processes with the specifications of each line.

II.3.2.4. Spaghetti diagram:

The Spaghetti diagram is a way to create a visualization of the actual flow.

This diagram is used essentially to track : the product flow (material physical flow), the information flow and the people flow. The purpose of this idea is to make quick and vigorous improvement with zero tolerance for waste, while implementing the project. [11]

II.3.2.5. Kaizen:

Kaizen is a Japanese concept that focuses on continuous improvement, when applied, can improve every function from manufacturing to assembly line work.

Kaizen aims to eliminate waste in all systems and organizations through improving standardized activities and processes. The continuous cycle of Kaizen activity has seven phases:

1. Identify an opportunity
2. Analyze the process
3. Develop an optimal solution
4. Implement the solution
5. Study the results
6. Standardize the solution
7. Plan for the future

Our aim by using Kaizen tool, is to coordinate efficiently all parts of the project, while making continuous improvement on the different operations. [12]

II.3.2.6. Heijunka:

Heijunka is a Japanese term that means literally “Leveling”, it’s a lean tool to level the line, minimize the chance of overburden and to reduce the unevenness in the production process. This tool helps to use the capacity and resources in the best way possible and to adapt the production react properly to demand changes. [13]

Implementing Heijunka will allow to:

- Organize the workflow based on customer demand.
- Meet the defined takt time.
- Avoid overburdens caused by demand spikes.
- Reduce inventory costs (avoiding overproduction).
- Insure an optimal running efficiency of our pull system.

II.3.3. Information flow pillar:

II.3.3.1. Push-Pull system:

A “pull system” is a lean manufacturing strategy used to reduce waste in the production process. This type of systems is used to meet the customer demand, so the process is based on replacing the components used when they are consumed. So the material is “pulled” whenever needed for immediate consumption, and so sold and making a profit.

Basically, the first steps of a pull system start from the customer demand, and based on the manner asked for production, the sequence and the method of production is installed.

On the other side, another production system can be used, a “push system”, where production is based on forecast demand. The units are “pushed” to the final assembly and so they need to be stored until they are needed. This system is based on prediction of customer demand. For that, we will need mass production for estimated future demand, which means that the production will remain in inventory for unlimited period. [14]

II.3.3.2. JIT: Just In Time:

By working with Just in Time (JIT), we are producing what is needed, when it is needed, and with the amount needed.

According to this production plan, we can eliminate the waste, inconsistencies and unreasonable requirements which bring a gain in improved productivity. [15]

This way, we will be able to concentrate our resources on fulfilling what will bring value, instead of building to large stocks.

II.3.3.3. JIS : Just In Sequence :

Just in sequence (JIS) is another inentory management strategy. With this production option, the component arrives to the assembly line at a specific moment when it’s needed, not before, which is the JIT case.

Each component arrives at the right time, at the right sequence and in adequate version.

Since the right part arrives at the right moment, the operator at the assembly line receives the correct part to assemble it with the final product, no need for warehousing or sorting.

II.3.3.4. Kanban:

Kanban means “Card” or “Signal”. The Kanban process is based on the idea of the supplier and customer. In the cycle, there is only a certain number of kanbans (boxes/trolleys with materials)

and the order for new material is made only when one box or trolley is finished in the production line. Therefore, kanban signals indicate demand, so overproduction is not allowed.

A kanban system usually performs efficiently and effectively in shop floor control when the demand is repetitive and stable, and the main factors that have very important influences on the Kanban process are: the factor of stability, the Replenishment Lead Time (RLT), flow degree and Work in Process (WIP).

Conclusion:

Gathering information related to the assembly process and the surrounded area, for a better understanding of the process and better analysis of the encountered problems, is essential to precede and understand the project. Literature review included different sources of information such as the observations, the discussions with the workers and the supervisors, the internal and external documents and articles and last studies in order to analyse the various methods and tools that will be used to solve the different issues.

After making our research and studies, and understanding the assembly process and the particularities of each version of HVAC units, we could understand that those differences are originally coming from the assembly of the pre-assembled parts, either in the sequence of assembly, or the differences in the small parts or materials. Each pre-assembled part is also compatible with a particular other part to get the final assembled part. Also this chapter was dedicated to a bibliographic study of the different lean manufacturing and management tools needed for the implementation of the assembly lines, the material flow study, and the information flow planning.

So our aim now is how to get the correct parts in the correct time when needed and precisely in the correct place where needed, by making the flow as organised and smooth as possible, while using the adequate tool for each issue.

Chapter III:

Project Scope:

Introduction:

In order to improve the efficiency and effectiveness of supply chain operations, it is necessary to ensure a well organized relationship between the flow of material and the flow of information and our resources.

This chapter will resume the practices realised and solutions brought, and also the expected future operations to be achieved in the next period.

This project also presents solutions and key elements to implement the different parts of the plan (assembly lines) for the assembly of air conditioning systems within the space allocated for the project. And so that the operation of implementing the assembly lines is achieved, 3 steps were fulfilled during my work at Valeo:

1. Studying the layouts and the principles of each assembly line:

Steps of assembly, material feeding and handling, relation between assembly lines, possible improvements, production volume, HVAC varieties…

➔ Loading: Time and quantity of batches delivered in each cycle.

2. Material flow management:

Analyzing the transportation means and network between all the pre-assembled parts (pre-assembled and finished assembled parts), analyzing the space allocated: buffering area, storage, material handling, time consumption, preparation of necessary materials for each stage of assembly, elimination the wastes (transportation, stocks…)

➔Rooting: Means of transportation of semi finished parts.

3. Information flow:

How / when / where / how often…

➔Scheduling: Planning of the beginning and end of tours/cycles.

III.1. Layout studies:

III.1.1. Current state analysis:

MRA2 project comes as a successor to MRA1 project for the production of HVAC units as part of the new Daimler business. MRA1 project is currently dedicated for the production of HVAC units for the Mercedes-Benz W205 (C-Class) and W213 (E-Class) with 2 variants each:

- Basic version: 2 zones. - Top version: 3 zones.

The MRA1 project started in 2014 as the first Daimler business, and is expected to touch the end by the year 2024. In the meanwhile, the start of project (SOP) of MRA2 will be initiated by the year 2020 and will last until 2030.

The biggest volume of demand is expected to exceed 900.000 (Biggest production volume ever in the plant). The figure 16 below shows the increase and decrease of the customer demand and the evolution of the 2 projects MRA1 and MRA2.

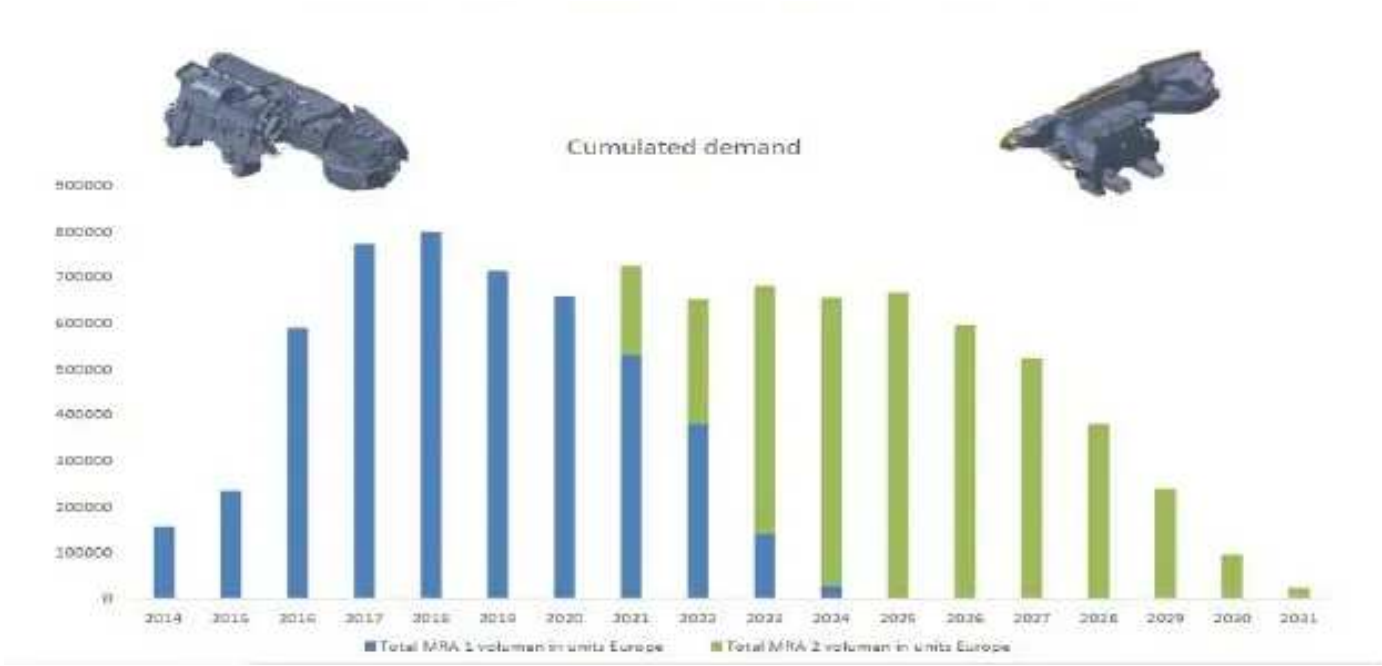


Figure 16. MRA1 & MRA2 production demand evolution.

The HVAC parts that will be totally manufactured in Bad Rodach site are air intake and evaporator pre-assembled parts, the 2 zones distribution housing pre-assembled parts and the finished HVACs, while the 4 zones distribution housing pre-assembled parts will be brought from another site (Rakovnik, Czech Republic).

III.1.2. Production growth:

The customer demand is expected to evolve, and the production in Bad Rodach is expected to increase until it hits a peak in 2025 with more than 910.000 HVAC units of all variants.

The table and diagram below (figure 17) show the production growth.

Year	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Total HVAC Production	1 346	225 877	451 750	817 057	899 667	910 124	853 186	751 581	561 490	265 184	94 026

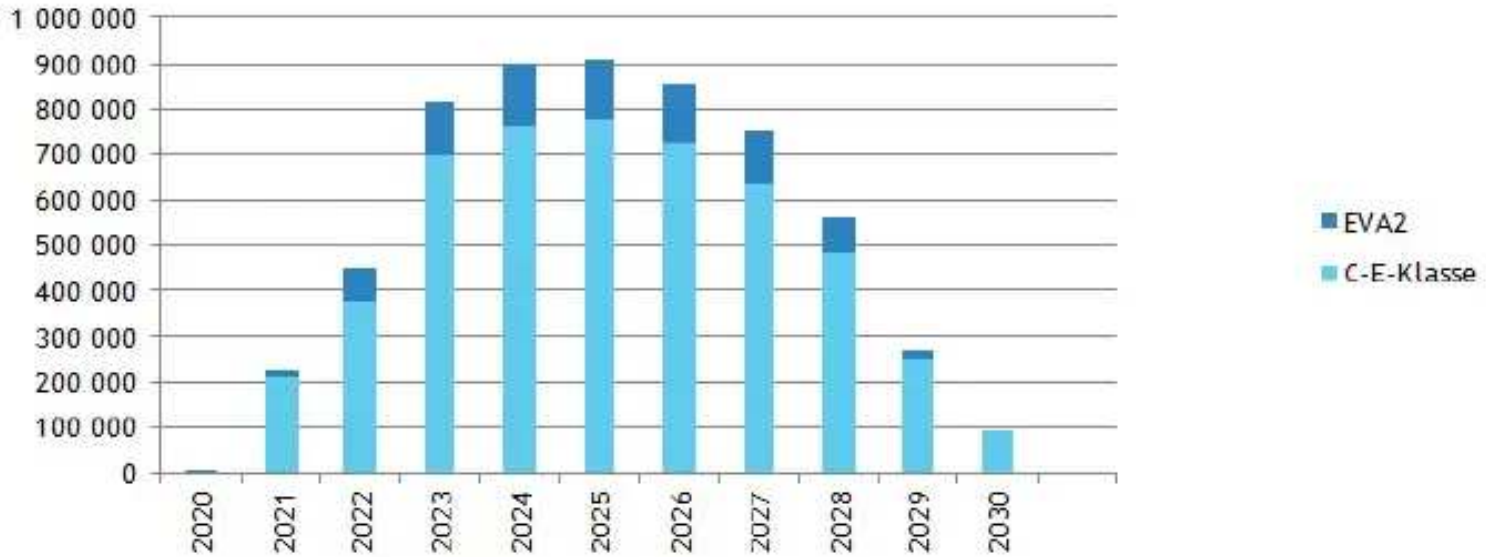


Figure 17. MRA2 production growth.

Based on the last table (Production evolution), our base (the layout) should be optimized and the production should be efficient and effective.

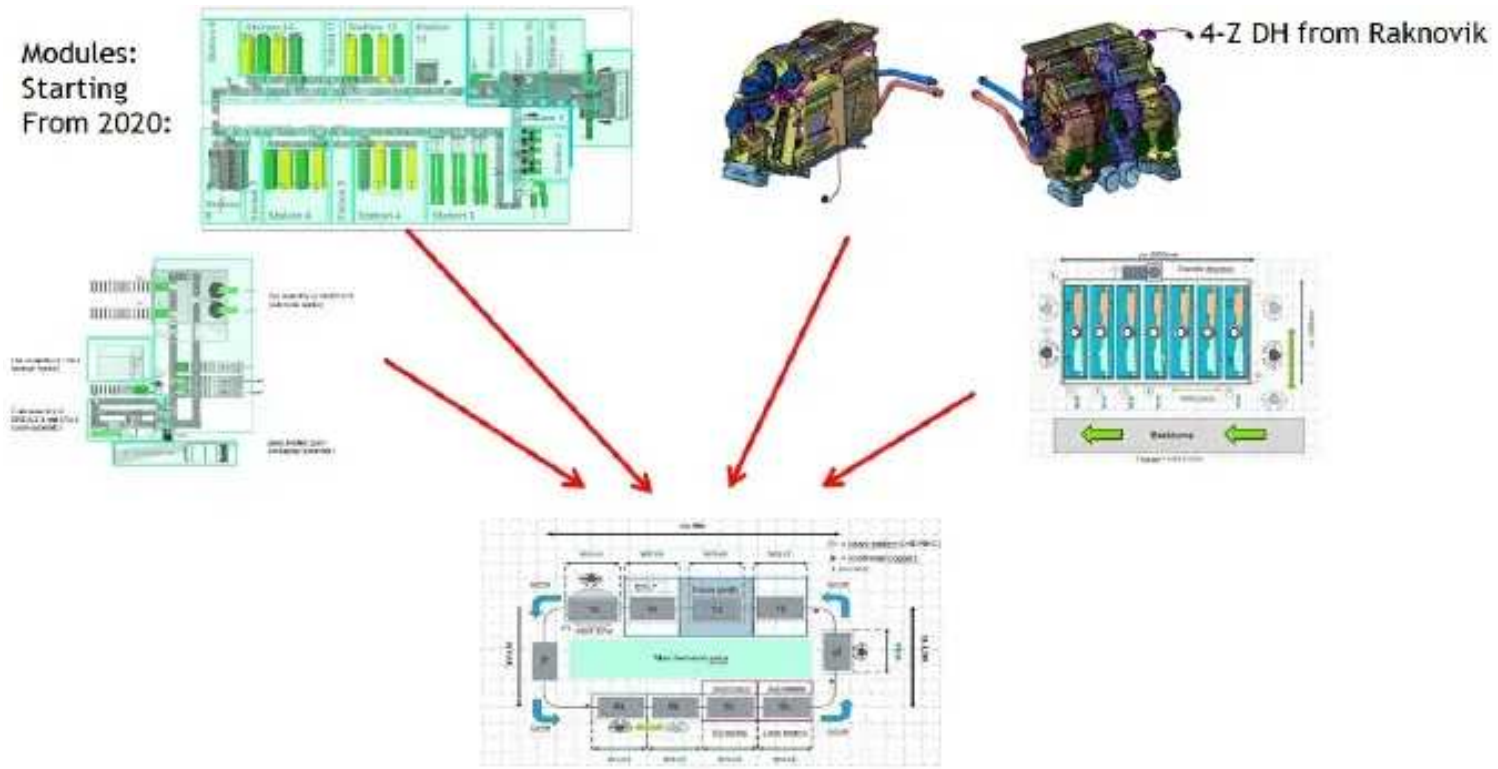


Figure 18. Assembly process and relation between assembly lines (2020 - 2021)

Starting from 2022, another production line will be introduced to double the output of HVAC units, so a decision was made by adding an evaporator module and a backbone module.

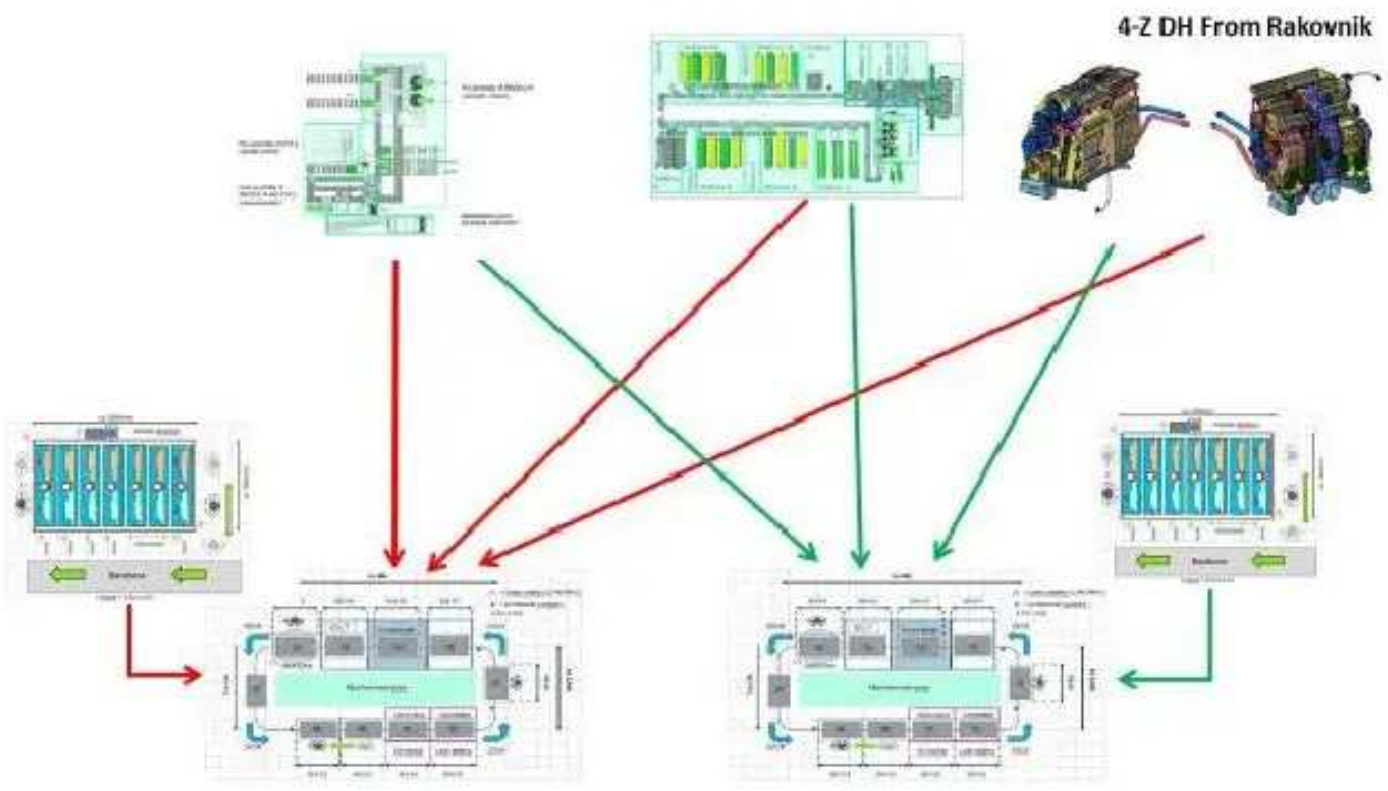


Figure 19. Assembly process and relation between assembly lines (2022 - 2030)

After finishing the assembly lines studies, the next step would be their implementation in the space allocated for the project, in the assembly department.

The annexe 7 and then the figure 20 show the layout and the position of each assembly line in real dimensions to have an idea how our layout should look like.

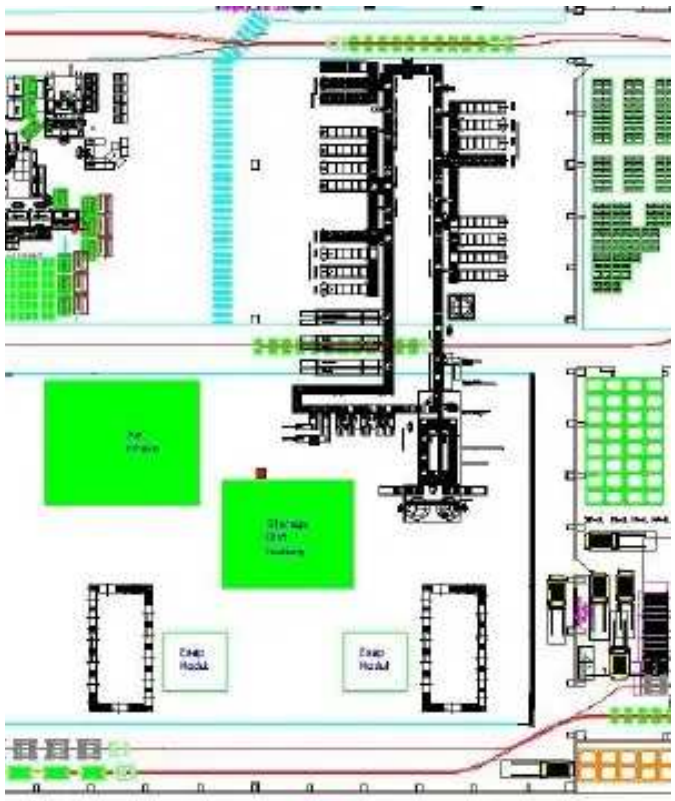


Figure 20. MRA2 Assembly lines implementation.

III.1.3. Conceptual design modification:

In order to adapt the layout of the assembly lines to the space allocated, and to have an optimized assembly process, some rearrangements and modifications on the conceptual design are needed with an effective use of resources, a minimum changeover time and a minimum of varieties.

III.1.3.1. Distribution Housing layout adaptation:

Some modifications on the assembly line and the feeding system can be realized, so that the line can fit in the upper space to gain more space without changing the assembly order.

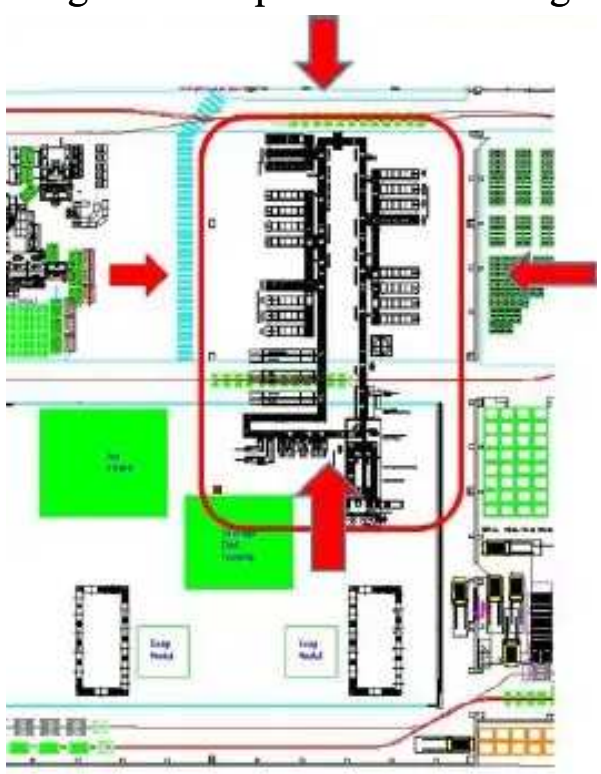


Figure 21. Distribution Housing module adaptation.

Some defects and optimization posiblities were noticed on the layout such as :

- **Over use of boxes:**

The first defect noticed and the first point to treat is the overuse of boxes for the housing parts in the stations 4,6,10 and 12.

The Distribution Housing is composed of the following 5 main pieces represented in figure 22 :

- 2 inner housings (Right / Left).

- 2 outer housings (right / Left).

- 1 Centerwall

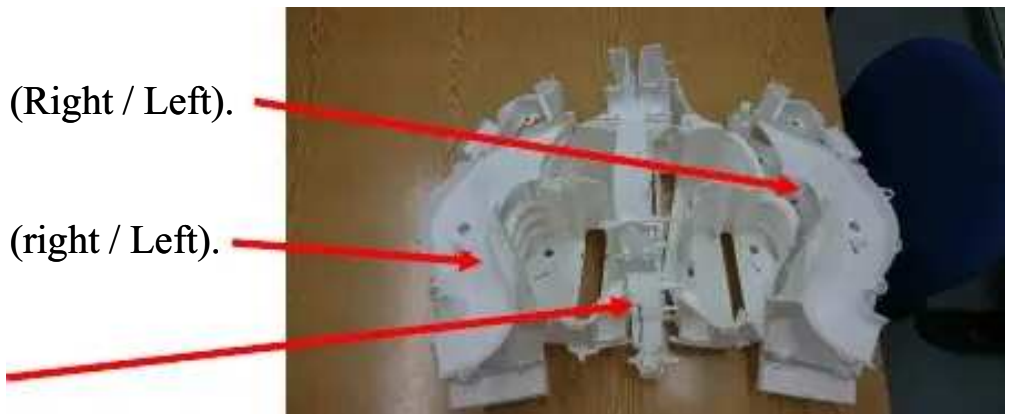


Figure 22. DH Housing parts.

The pieces are fed to the assembly line in 2,5 boxes for each pre-assembled DH.

If we take as an example the year 2025 (peak year of production), we will be producing 147 DH/Hour, which means 368 Boxes / Hour (2940 Boxes / Shift – 3220 Boxes / Day).

The dimension of 1 Box is: 0,6mm x 0,4mm x 0,235mm = 0,0564m³.

So in 1 hour, we will have a volume of 0,0564 x 368 = 20,75 m³ allocated by the boxes only.

With the advancement of the project, the use of boxes and the space allocated will be huge and eventually, it will be hard to handle it (storage, picking, transporting): a problem to solve.

- **Reducing the long distances:**

Some points in the layout represent non-used long conveying distances. Rotary index table and changing the form of the line to fit in the space are mandatory as solutions to this issue, by the use of circular tables (or semi-circular) rotating to advance parts between stations located along its perimeter, since a rotary table is restricted to circular transfers with a single compact work center. This solution can be used in the pre-fixation of small parts station (station 2).

III.1.3.2. Air intake layout optimization:

Some modifications are made on the AI assembly line to ensure a capable material feeding in less space with more ergonomic operation.

In order to accelerate the operations of assembly and to make them more efficient, a rotary table will fix this issue. The figure 23 below shows our new Air Intake assembly line, using a rotary table, and a line adaptation to the different varieties.

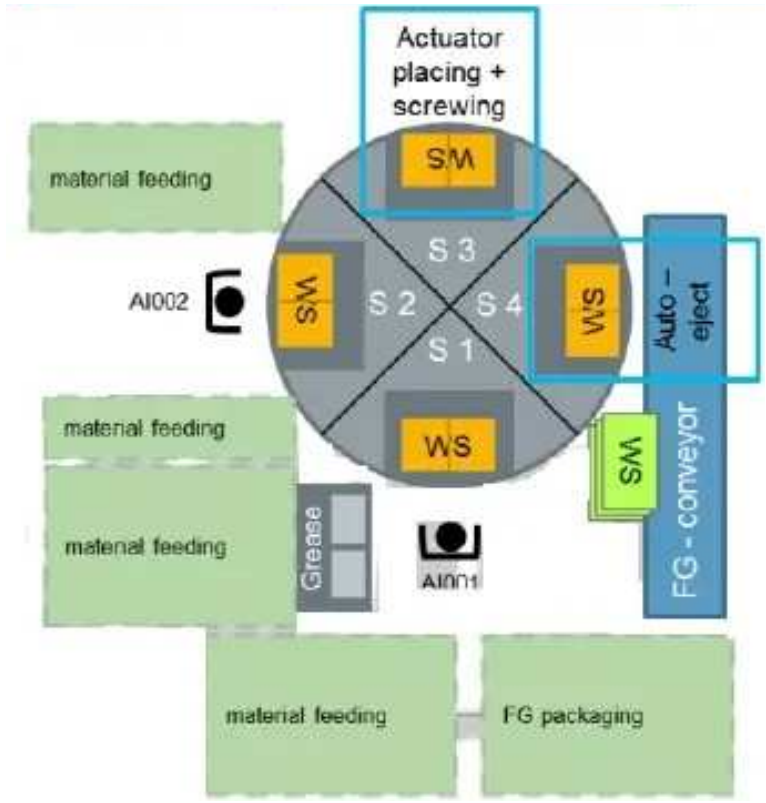


Figure 23. Layout of Air Intake module for MRA2 and EVA2

III.1.4. Work In Process:

Work In Process (WIP) is the set of semi finished products located on the factory floor. These goods are located between raw materials and finished products in the production process flow [16]. In our case, work in Process is represented respectively by the 2-Z and 4-Z Distribution Housing, Air Intake and Evaporator semi finished parts. So, every pre-assembled part, due to the time lag between the modules and the customer demand fluctuation, will need to be stored in a buffer area. Calculations were made for each variety and collected the table 4, to have an idea about our production of finished HVACs, and also the correspondent varieties.

Year	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Total HVAC Production	1 346	225 [illegible]	451 750	847 [illegible]	899 66[illegible]	910 124	953 [illegible]	751 [illegible]	561 [illegible]	266 154	94 [illegible]
Production/Day	5	892,794466	1799,8008	3281,35341	3584,33068	3640,496	3385,65873	2970,67589	2254,97992	1060,736	376,104
Production/Shift		297,598155	599,933599	1093,78447	1194,77689	1213,49867	1128,55291	990,225296	751,659973	353,578667	125,368
Production/Hour		40,7668706	82,1826848	149,833489	163,668067	166,232694	154,596289	135,647301	102,96712	48,4354338	17,1736986
Round		41	83	150	164	167	155	136	103	49	18
For Production Bad Rodage	1 [illegible]	243 [illegible]	415 [illegible]	387 [illegible]	403 [illegible]	800 242	790 [illegible]	696 [illegible]	885 [illegible]	721 [illegible]	89 [illegible]
Production/Day	5	836,980237	1653,42629	3002,80321	3200,26295	3200,968	2976,90476	2602,01581	1948,12048	886,08	277,24
Production/Shift		278,993412	551,142098	1000,9344	1066,75432	1066,98933	992,301587	867,338603	649,373494	295,36	92,4133333
Production/Hour		38,2182757	75,4989176	137,114302	146,130728	146,162922	135,931724	118,813507	88,9552731	40,460274	12,6593607
Round		39	76	138	147	147	136	119	89	41	13
For stored from Inventory	77	14 [illegible]	36 [illegible]	65 [illegible]	96 [illegible]	109 883	163 [illegible]	93 [illegible]	96 [illegible]	43 [illegible]	26 [illegible]
Storage/Day	0	55,8181818	146,374502	278,550201	384,067729	439,532	408,753968	368,660079	306,855422	174,66	98,864
Storage/Shift		18,6060606	48,7915007	92,8500669	128,022576	146,510667	136,251323	122,886693	102,285141	58,22	32,9546667
Storage/Hour		2,54877543	6,68376721	12,7191873	17,5373392	20,0699543	18,6645648	16,8337936	14,0116631	7,97534247	4,5143379
Air Production	1 346	225 [illegible]	451 750	847 [illegible]	899 66[illegible]	910 124	953 [illegible]	751 [illegible]	561 [illegible]	266 154	94 [illegible]
Production/Day	5	892,794466	1799,8008	3281,35341	3584,33068	3640,496	3385,65873	2970,67589	2254,97992	1060,736	376,104
Production/Shift		297,598155	599,933599	1093,78447	1194,77689	1213,49867	1128,55291	990,225296	751,659973	353,578667	125,368
Production/Hour		40,7668706	82,1826848	149,833489	163,668067	166,232694	154,596289	135,647301	102,96712	48,4354338	17,1736986
Round		41	83	150	164	167	155	136	103	49	18

Table 4. HVAC, DH, AI expected production : WIP.

III.1.5. New layout:

After introducing different modifications on the layout, and discussions with the suppliers, a new version of our assembly line area is presented affording more adapted modules to the space allocated. The figure in the appendix 8 shows the new layout.

« flow management between the lines: internal Material flow":

III.2. Material Flow:

The main topic of my internship articulates around material flow implementation in the production area and giving solutions for a smooth movement between the lines. Our target is to ensure this critical operation and several steps are to be followed to attend our objective.

In this part of the chapter, I will be showing the tools and the multiple ideas to find the best option depending on its feasibility and the criterias of the investments.

III.2.1. Material flow management between assembly lines:

In order to develop an assembly system, it's necessary to consider all aspects of production system. And material / information flow management within the company is one of the most important topics while planning the implementation of assembly lines.

Material flow management refers essentially to the planning and allocation of necessary resources to each production, assembly unit and module during the planned time schedule, and so, the main purpose is to minimize the steps and stops, distance and buffering spaces that an item goes through while moving between the lines.

In our case where production is divided between separate units or modules, transportation means of products to and/or from the lines is very important, in terms of capacity, space, number and compatibility of means of transport, so various transportation modes could be applied for the process. And while implementing the plan, the management system gets more complicated due to the physical state of the materials and intermediate products (WIP), size, dimension, form, varieties of the same product…

So our aim is to make the flow as smooth and optimized as possible.

III.2.2. Material flow: MRA2 Project:

This paragraph describes the material flow as expected to be realized.

First, I will present an overview about the layout implementation and localization of the intermediate storage. Then, calculations are made for a prediction of the production volume evolution during the project lifecycle to have an idea about the workflow and WIP evolution for a clearer image about the space management and time scheduling in the area allocated for storage and production. Next, a spaghetti diagram is showing precisely how the material moves

in the assembly area. To finish, different ideas about storage, packaging and transporting options are mentioned based on space, ergonomics and investment feasibility.

III.2.2.1. Layout plan overview:

After studying different assembly lines for automotive assembly in the company, different steps and solutions for the transportation network and the space management were introduced to get an optimized material flow for assembly.

The main points that assembly lines focus on, are time consumption, regulation of materials flow and preparation of the necessary materials for each module exactly when it is needed.
Based on this, it is extremely necessary to prevent the increase of unnecessary temporary stocks in the process of production and to decrease transportation activities in order to reduce the assembly's indirect costs (storage and transportation are among activities that bring no added value for the system, but add very high costs.).
For observation of the transportation of pre-assembled parts and final assembly, we must have

a real time information about the state of the different steps of assembly : stocks of parts, stocks of intermediate warehouse (buffer area) and production volume.

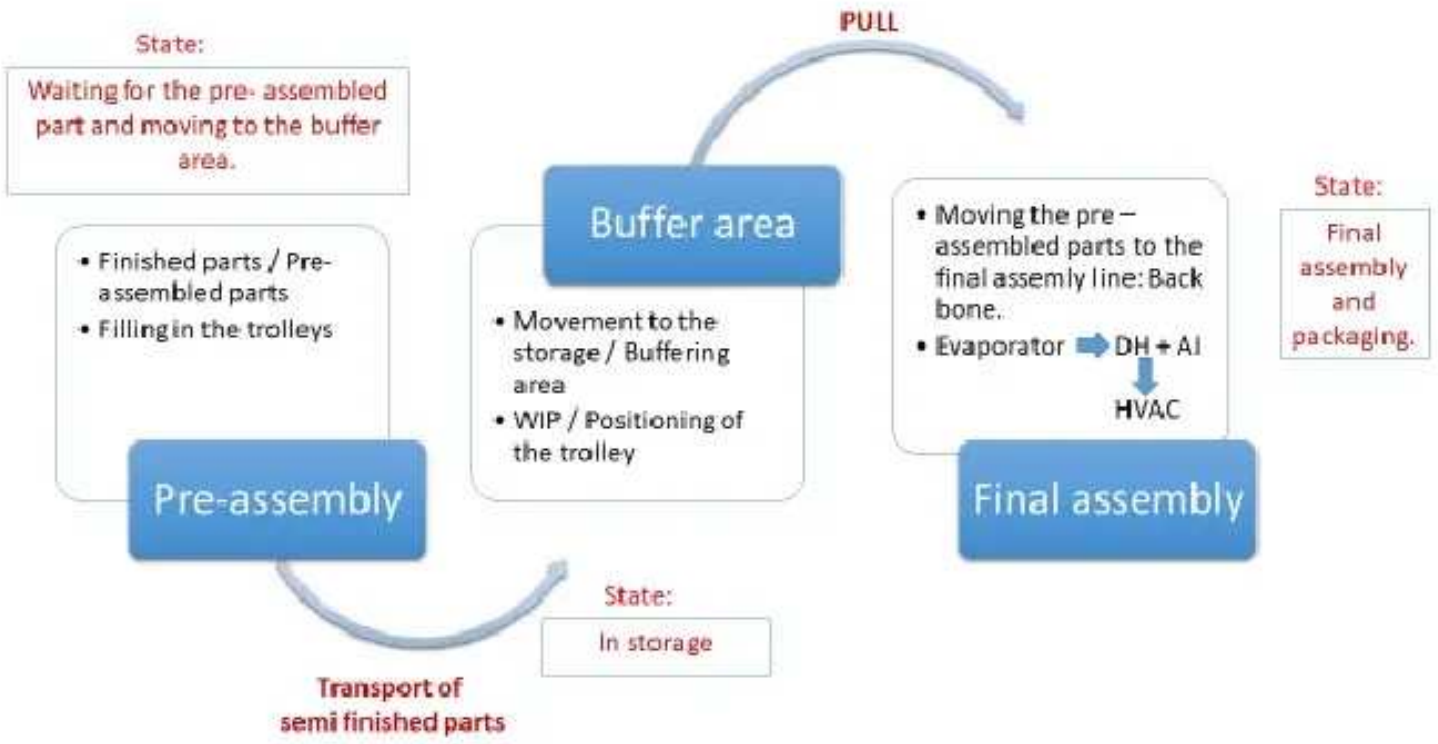


Figure 24. Overview of the movement of parts within the assembly area.

All the steps of assembly and relation between the parts were studied, and in order to ensure an effective response to the customer demand, a production volume study was made so we can have an idea about the volume, the space needed and a time table for production for an overall

organization. So, to understand the process physical flow, an Ishikawa diagram was elaborated showing the different steps of assembly of HVAC systems in the appendix 9.

III.2.2.2. Packaging: Loading and unloading options:

A packaging option is needed to transport the distribution housing pre-assembled parts and the air intake pre-assembled part. The objective at this point is to find the best trolley / packaging option for each part in order to transport it, first from the assembly line (modules) to the buffer area, and then from the buffer area to the backbone.

I compared the capacity and the space needed by using different types of trolleys to find the best transporting and packaging option.

The figures 25, 26 and 27 and the tables 5 and 6 show different options afforded with a comparison in term of space needed and capacity to handle the Air Intake and Distribution Housing pre-assembled parts.

III.2.2.2.1. Distribution housing packaging:

Trolley used:	Length (m)	Width (m)	Height (m)	Surface (m^2)	Volume (m^3)	Capacity
5 shelves White trolley	0,88	0,6	2,09	0,528	1,10352	20 DH
4 shelves White trolley	0,88	0,6	1,8	0,528	0,9504	16 DH
Hordenwagen (White trolley)	1,51	1,08	1,59	1,6308	2,592972	24 DH

Table 5. Trolleys specifications for DH packaging.



Hordenwagen	5 shelves White trolley	4 shelves White trolley

Figure 25. Trolleys options for DH packaging

III.2.2.2.2. Air Intake packaging:

Trolley used:	Length (m)	Width (m)	Height (m)	Surface (m^2)	Volume (m^3)	Capacity
Standard Green Trolley	1,46	0,88	1,82	1,2848	2,338336	32
White Shelves trolley	0,88	0,6	1,8	0,528	0,9504	28

Table 6. Trolleys specifications for AI packaging.



Standard Green Trolley	White Shelves trolley

Figure 26. Trolleys options for AI packaging

Based on the caracteristics of each trolley (capacity, cost of investment, space allocated and quality issues), the final choice for the packaging options for air intake and distribution housing was made for the white shelves trolley: this type of trolleys is effecient and effective in capacity and space, doesn't need an investment and is suitable for our parts.

III.2.2.3. Space of storage:

Once the suitable trolleys were found, to store and transport the DH and AI parts, a space study is mandatory to reserve the space covered in the buffer area based on the daily production in order to manage the material flow the best way possible.

-DH Production: after choosing the final packaging, the table of DH production and space calculation for buffer area (Appendix 10) gives an idea about the production volume in Bad Rodach (2-Z) and the parts brought from Rakovnik (4-Z), as well as the space needed for the storage (per day / per shift / per hour).

-AI production: Air Intake pre-assembled parts are totally produced in Bad Rodach site. They include 4 different varieties. The calculations are collected in the appendix 11.

III.2.2.4. Global Overview of the flow between the modules:

In this paragraph, I will be showing a detailed general overview of the material flow by analyzing and planning of the flow from pre-assembly lines to the backbone (final assembly) including the material handling.

A Spaghetti diagram in the figure 26 and also in the appendix 12 were drawn, showing precisely how the pre-assembled parts should be transported. This will help to understand the flow and trace the path of each item. It was also beneficial to see where the difficulties could appear and pay attention to opportunities.

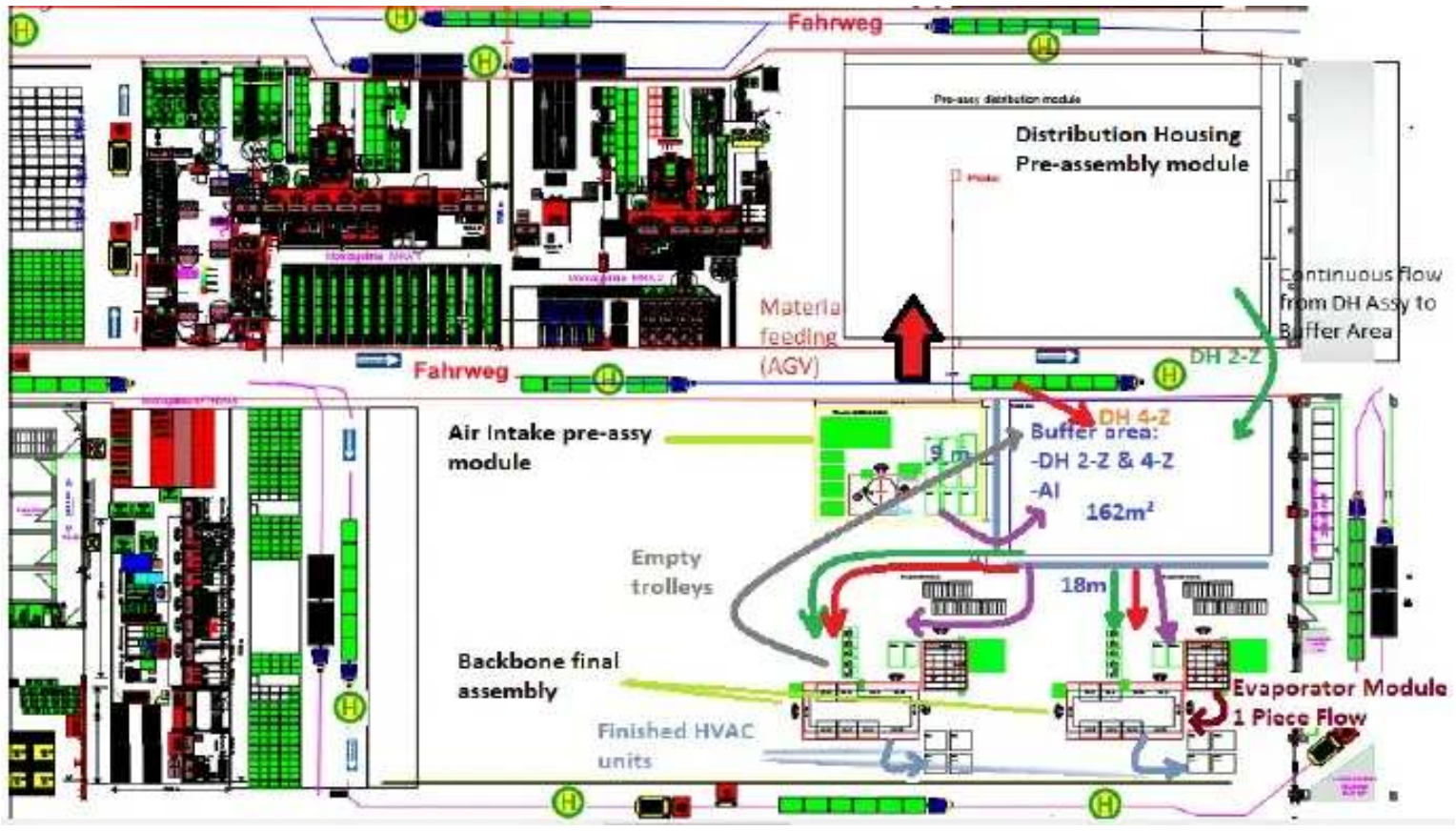


Figure 27. Spaghetti Diagram.

III.2.2.4.1. From Distribution Housing Assembly line to storage:

Every 18s, the cycle time of distribution housing module, a distribution housing pre-assembled part is transported to the buffer area.

The idea is to change the classic manner of transportation using trolleys, instead we can use an overhead conveyor system allowing to take the parts to the other side of the street with no need of extra movements. An operator will be filling the empty trolleys with the DH pre-assembled parts coming by the overhead conveyor (20 DH / Trolley). Once the trolley is full, it will be placed in the correspondent variant queue, ready to be transported later to the back bone

➔ Continuous feeding of parts to the trolleys.

The condition to respect when installing the overhead conveyors is it has to be higher than 3m because of the passage of the AGVs regularly in the guideway “einbahnstrasse”.

The table “DH Production and space calculation for buffer area” (Appendix 10) shows the estimated distribution housing production volume and the needed trolleys quantity and space allocated in the buffer area (per year / per shift/ per hour).

Example: in the peak year of production: 2025, the total DH production in Bad Rodach will attend 800.242 DH, which is equivalent to 3200 DH/Day and so the hourly production is estimated by 147 DH pre-assembled parts.

The space and trolleys needed will be discussed in the paragraph “Buffer area”.

III.2.2.4.2. From the storage warehouse to the distribution housing 4-Z Storage:

The 4-Z Distribution Housing parts are pre-assembled in another Valeo plant: “Rakovnik, Czech Republic” plant, and are brought ready to be assembled with the rest of the HVAC parts. So the DH 4-Z Parts will be brought from the local storage warehouse to the buffer area in the assembly department to be stored with the 2-Z DH locally pre-assembled (we use the same trolleys used for DH 2-Z (20 DH/Trolley)). The table “DH Production and space calculation for buffer area” (Appendix 10) gives us an idea about the DH 4-Z Production, the number of trolleys needed and the space allocated in the Buffer Area.

Example: In the peak year (2025), the volume of 4-Z DH will attend nearly 110.000 DH from Rakovnik, which means:

- 440 DH/Day ⇒ 22 Trolleys / Day.
- 147 DH/Shift ⇒ 8 Trolleys / Shift.
- 21 DH / Hour ⇒ ~ 1 Trolley / Hour.

The most efficient way to bring the trolleys from the storage warehouse would be to bring them 3 times/ day, every beginning of a shift with an order of 8 Trolleys / Shift (2025).

III.2.2.4.3. From Air Intake Assembly line to AI Storage:

The pre-assembled Air Intake parts are filled in the white shelves trolleys (28AI/Trolley), every 18s. Every time a trolley is full, it is placed in the queue line in the buffer area waiting to be pulled to the Back Bone when the variety is needed.

The table “AI Production and space allocated for the buffer area” (Appendix 11) shows the evolution of the production volume and the space allocated for storage.

III.2.2.4.4. Consistency of production of Air Intake and Distribution Housing parts:

Both AI and DH Modules have the same cycle time.

Air Intake presents 4 different varieties:
MRA2 ⇒ Left Side / Right Side. / EVA2 ⇒ Left Side / Right Side.

Distribution Housing introduces 8 different varieties:

- MRA2: ⇒ Left Side: 2 zones / 4 zones & Right Side: 2 zones / 4 zones.
- EVA2: ⇒ Left Side: 2 zones / 4 zones & Right Side: 2 zones / 4 zones.

In order to decrease the number of changeovers and for less wasted time, a synchronization between the production of air intake and distribution housing is mandatory.

The table « Air Intake and correspondent Distribution Housing Varieties » (Appendix 13) shows how the production of the same variety of AI is compatible with different varieties of Distribution Housing parts. The table shows also a detailed volume of production of all varieties per year and per day and the evolution in numbers of each version.

III.2.2.4.5. Buffer area:

The buffer area is the space dedicated for the storage of the pre-assembled distribution housing (2 zones and 4 zones) and air intake. The repartition of the space is based on the quantity of the parts, the number of trolleys, the varieties type, the order of production, and the flow of the different varieties.

The space is divided into High and Low runners and for each variant a specified line will be dedicated for storage. The tables “Distribution Housing Production” and “Air Intake and correspondent Distribution Housing Varieties” (Appendixes 10 and 13) show an average space of buffer area and the trolleys for each part and variety during the period of the project and the possibility to divide quantity and space per day, shift or hour.

Example: During the peak year: 2025:

Distribution Housing 2-zones:	Distribution Housing 4-zones:
3200 DH / Day ⇒ 161 Trolleys ⇒ 86,94 m²	440 DH / Day ⇒ 22 Trolleys ⇒ 11,88 m²
1067 DH / Shift ⇒ 54 Trolleys ⇒ 29,16 m²	147 DH / Shift ⇒ 8 Trolleys ⇒ 4,32 m²
147 DH / Hour ⇒ 8 Trolleys ⇒ 4,32 m²	20 DH / Hour ⇒ 1 Trolleys ⇒ 0,54 m²

All calculations made, the space reserved for the buffering is around 162 m² (9m x 18m).

Theoretically, we will be having one trolley full of DH every 360s (18s x 20), but this filling time depends on the period and efficiency of production (for example, we expect to have 1 full trolley of 2-Z DH every 450s = 7,5 min (3600s / 8)).

The diagram in figure 28 describes the surface covered in square meter with the trolleys in the buffer area and its evolution in time.

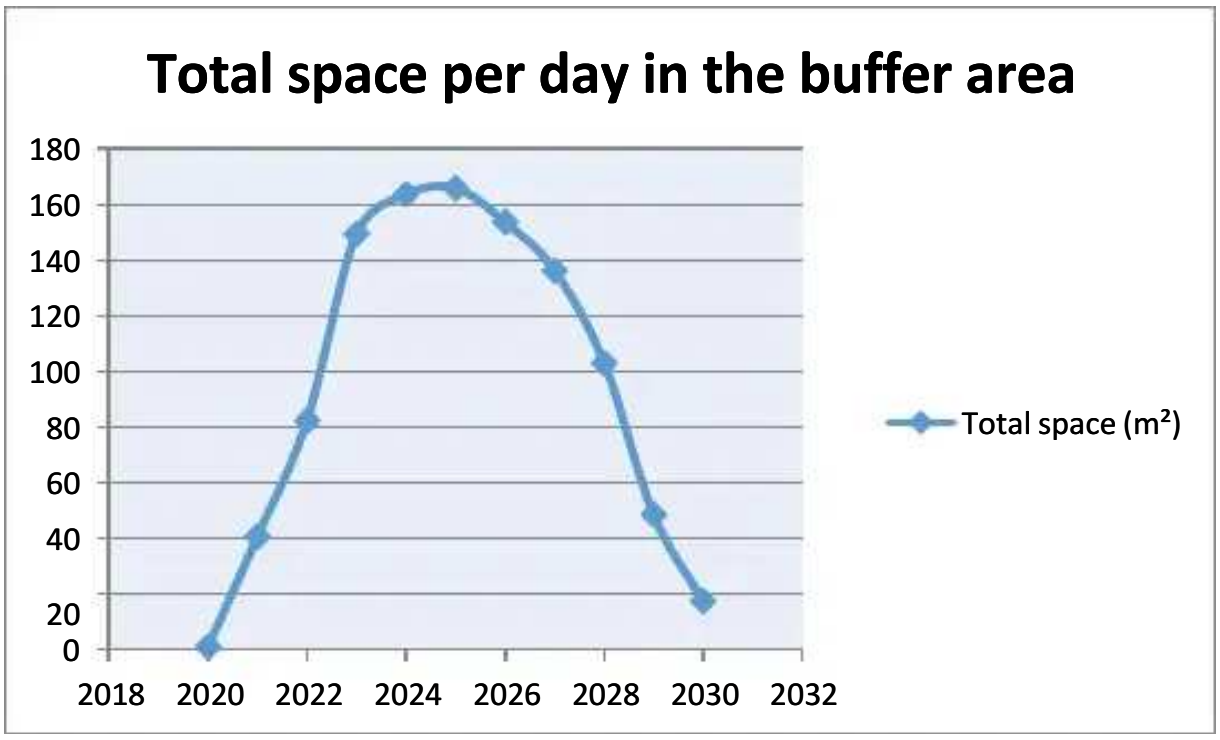


Figure 28. Total daily surface covered with full trolleys in the buffer area (2020-2030)

III.2.2.4.6. From storage to backbone:

The backbone has a cycle time of 38s (2 x the cycle time of DH and AI) and the same cycle time of the evaporator line. So, everytime an evaporator pre-assembled part is finished (38s), it is placed by the operator on the backbone module, to finish the assembly of the HVAC: « One piece flow ».

The idea includes the presence of the correct version of distribution housing and air intake pre-assembled parts, to the final assembly lines (both backbones) when needed.

Bringing the trolleys should be as mentioned below, using a Pull principle.

To achieve that, we need further methods and actions:

- Leveling our production: Heijunka / SMED: Cover the fluctuations.
- Pull production and Pull deliveries.
- Just In Sequence: many variants ⇒ Delivering in the right sequence and time.
- Focusing on the high runners: large quantity ➔ less fluctuations.
- KANBAN.

JIT VS. JIS :

JIT and JIS react differently to the production process. Each of them has its strategy in the organisation process of production as described in the figure 29.

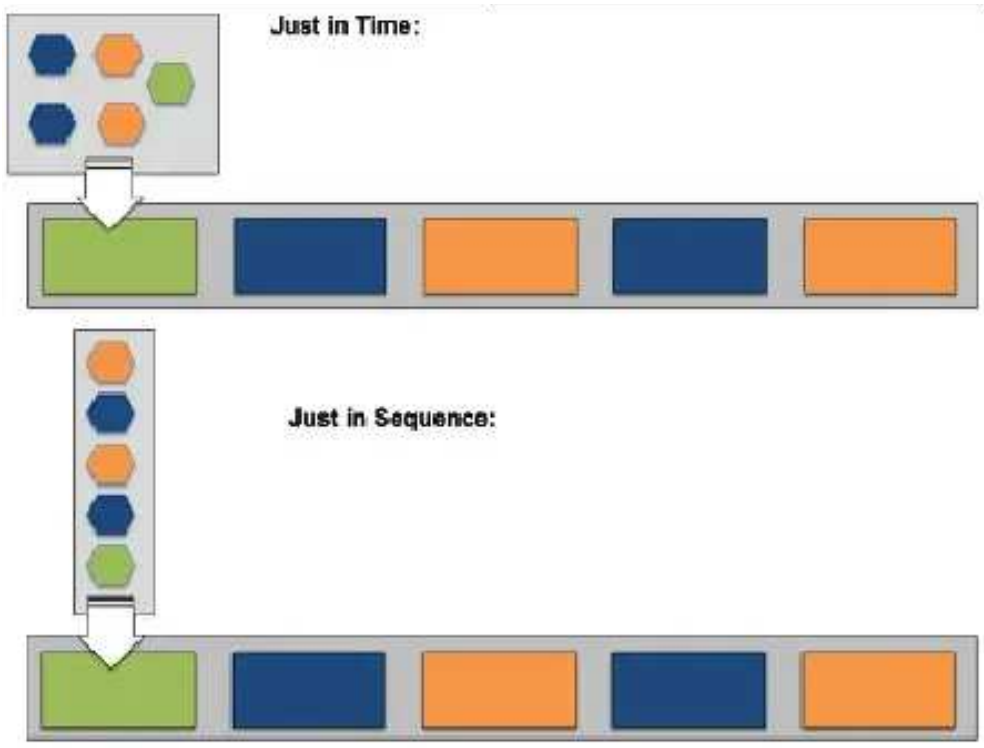


Figure 29. Just In Time and Just In Sequence in production process.

The differences in terms of caracteristics between the two strategies of production are listed in the table below.

Caracteristics	**JIT**	**JIS**
Safety Buffer	Inefficient.	Highly inefficient.
Module specification	Low.	Maximal.
Requirements for IT system	Integrated.	Fully integrated.
Stability of sequence.	Irrelevant.	Absolute necessity.
Downturn reaction times.	Hours / Days.	Hours / minutes.
Switching costs.	High.	Maximal.

Table 7. Difference between the caracteristics of JIT and JIS.

Just In Sequence requires a high level of synchronization between production process of suppliers and customers.

III.2.2.5. Classification of the buffer area: High and low runners:

The finished goods are classified in our buffer area as low and high runners. This classification will insure a minimum storage time and space.

Our goods are distributed as A-parts, B-parts and C-parts, so the A-parts are high runners and C-parts are low runners. The table of the varieties of pre-assembled parts was elaborated to show, from one hand, the air intake varieties compatible with the correspondent distribution housing varieties, and from the other hand, the production volume evolution of each variety per year and per day (Appendix 13), which gives an idea about our high and low runners.

The diagram ABC (figure 30) method shows the classification of the different parts.

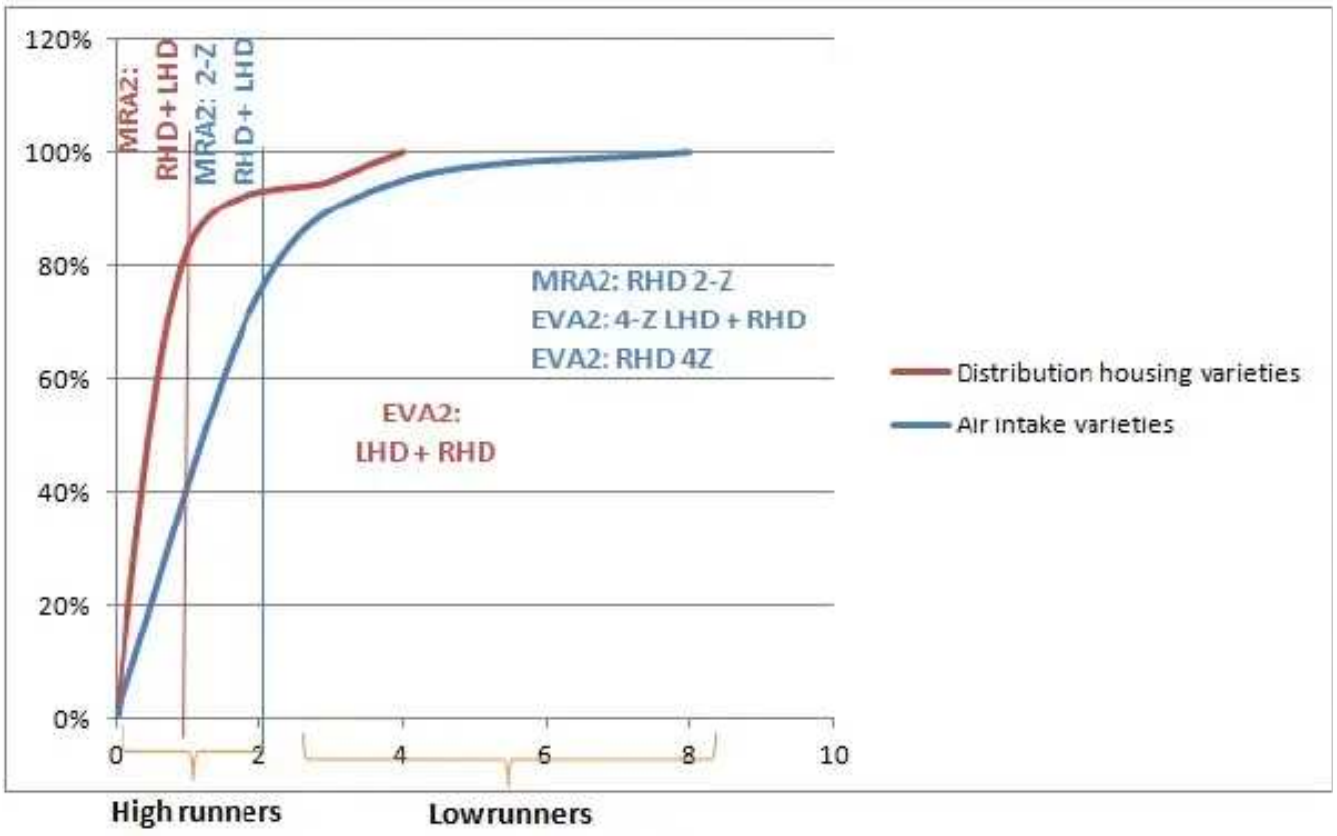


Figure 30. Diagram ABC : High and low runners classification.

- **High runners: A: 75% of the total production:**

  Our distribution housing high runners are the MRA2 LHD 2-Z (2131 DH/day) and MRA2 RHD 2-Z (626 DH/day).

  Our Air Intake high runners are the MRA2 LHD (2414 AI) and MRA2 RHD (694 AI) varieties.

- **Low runners: C: 10% of total production:**

  Our distribution housing low runners includes 4 varieties: MRA2 RHD 4-Z (76 DH/day), EVA2 LHD 4-Z (70DH), EVA2 RHD 2-Z (59DH/day) and EVA2 RHD 4-Z (10DH/day).

  Our air intake low runners are the EVA2 varieties: LHD (464 AI/day) and RHD (70 AI/day).

III.2.2.6. Backbone output:

Our output, the finally assembled HVAC unit, will be produced starting from 2022 until 2030, with two separated Backbones related to two evaporator modules, as shown in the new layout. As a matter of logistics and production growth, we were having two options of production, each one of them will has a different impact on the pre-assembled parts (Distribution housing, Air Intake, Evaporator) process.

Option 1: We produce the same variety in the 2 backbones at the same time.

Option 2: We produce separately different varieties for each backbone.

Two SWOT matrixes were elaborated to show the differences and the impact as well as the advantages and disadvantages of each option of production.

Option 1: Two Backbones, same variety:

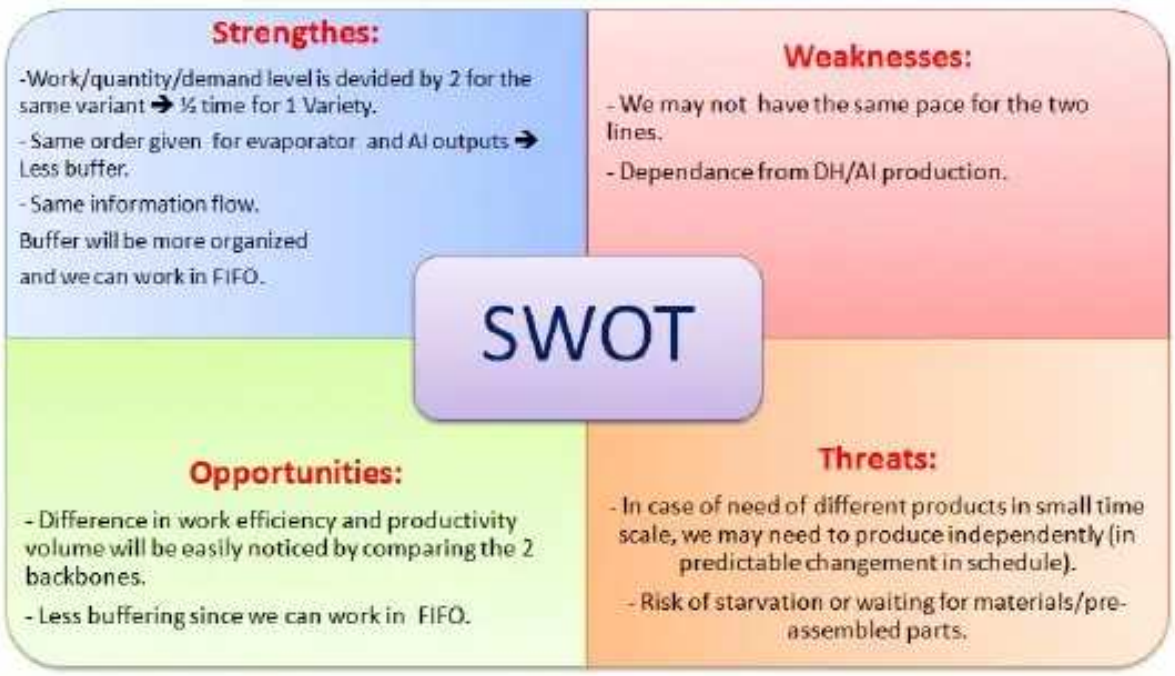


Figure 31. SWOT Matrix for the production of the same variety in the two backbones.

Option 2: Two Backbones, different varieties:

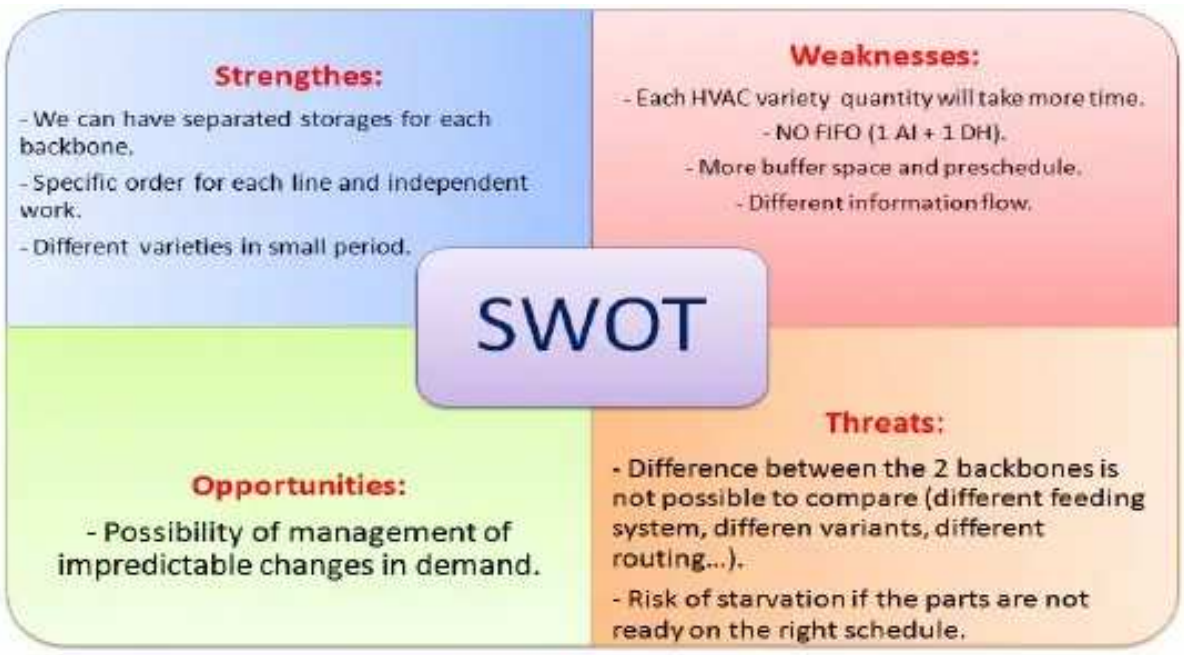


Figure 32. SWOT Matrix for the production of different varieties in the two backbones.

The comparison between the advantages and disadvantages of each type of HVAC production brings as conclusion that the production of the same variety in the two backbones, and so having the same output at the finish line, is more advantageous than a different production of varieties.

III.3. Information Flow:

The customer would be communicating his demands of the product volume and variety early before production (4 to 6 months forecast) on a daily electronic request.

Managing the different operations inside the company such as integrated planning, controlling, production and material and information flow is very important for the life of the company. So, in order to manage the manufacturing, JIT, JIS and Kanban, E-Kanban and EBTT are helping, among other tools, to minimize the costs and maximize productivity, so that quality is insured and the storage is as low as possible.

III.3.1. Kanban:

Kanban system is considered as one of the most important components of Toyota Production System (TPS), as it's a simple and effective tool to accomplish the Pull concept of Lean.

Kanban's core is based on the following principles:

- Work visualization: Through Kanban system, we can observe the flow of work moving by creating a visual model of work and work flow.
- Work In Process limitation: By limiting the work in process volume (unfinished parts) we can reduce the time that an item takes to travel through the kanban system. This is a way to avoid the problems caused by task switching and reduce the need to constantly re-prioritize items.
- Focusing on flow: The optimization of the kanban system in order to improve the flow of work, analyzing the flow and getting leading indicators of future problems by analyzing the flow of work and limiting the WIP.
- Continuous improvement: The culture of continuous improvement is set, once the Kanban system is in place, by tracking the flow, quality, lead time… ➔ Experiments and analysis may change the system to improve the team's effectiveness.

III.3.2. Work in Process limitation:

After understanding how the flow runs, the next step would be learning how to optimize the workflow. The most effective way to optimize flow in kanban, is by limiting WIP. WIP is communicated as numbers and by controlling it we can manage capacity, achieve team goals by encouraging the practice systems thinking and identify opportunities for process improvement. WIP limits enable to slow down enough, and work intentionally enough, to create space for growth.

III.3.3. E-Kanban:

E-Kanban is a process based on the internet use application. It's a way to transfer the kanban signal automatically via a barcode scan from downstream to the upstream assembly lines (real time exchange of informations.).

This very quickly changing environment demanding high adaptability asks also for very flexible way of material ordering.

III.3.4. Kanban Vs. E-Kanban:

Kanban is now a way to implement smoothly agile and lean management methods within companies around the world.

However, under the global competition in the recent years, the increased product variety and physical distance among suppliers in supply chains, the complexity of Kanban applications and mistakes resulted are increasing, such as delays, lost cards, and incorrect kanban deliveries.

Besides, traditional Kanban lacks the ability to track and monitor the physical cards, so by using electronic Kanban (e-kanban), the existing problems with traditional Kanban can be solved.

The table 8 is a comparison between classic Kanban card and electronic Kanban methodology.

	Kanban Card.	E-Kanban
Control of ordered material level.	X	X
Easier and faster ordering of material.	X	X
Easy work for material handlers.	X	X
Visualization of problems in production.	X	
Setting of priorities.	X	
Regulation and optimization of stock.	X	X
Simplification of production planning.	X	X
Improves the flexibility oft he production line.	X	
Works with high amount of materials.		X
Long distances between stations.		X
Quick and precise info.		X
Simple and easy handling of thousands of Kanban loops and cards		X

Table 8. Kanban Card Vs. E-Kanban.

The comparison shows that e-kanban is much more advantageous than the classic Kanban cards.

III.3.5. E-Kanban process:

When the operator in the “Back bone” needs the correct parts he scans the barcode in the Kanban card. Once the barcode is scanned, an order is generated in SAP and the status changes from “empty” to “in process”.

The barcode contains the different information concerning the parts (type of the variety, the quantity, position, the source and destination of the material).

When the material arrives to the final assembly line, it is checked by the operator whether it is the correct part with the correct amount, and then he scans the barcode on the same kanban card with which he ordered the modules, and the status changes to “full”, and the card is ready to be used again.

Prospects:

New ideas for storage and material flow studies:

After a study of the material flow analysis, we can notice that the buffer in stock area is covering an important space with the current tools of storage, which is considered as a ´Muda´ (waste). This huge buffer area will lead to difficulties to react to changements and to adapt the assembly lines to fulfill the demand, without mentioning the overburden and the traffic jam.

With high inventory and production rates, especially in the peak years of production (2024-2026), it is mandatory to minimize the buffer area, or to find a better solution for storage.

At first glance, the idea was about ‘Cubic’ utilization of the area of storage: Air storage (cubic matrix) which will create more space of inventory for the same surface.



Figure 33. Cubic storage: Air exploitation.

Idea 1: Auto store solution: Currently under studies:

This solution will reduce the storage surface by creating more volumic space, and with automatic handling and transportation of the parts from and to the assembly lines.

The system is composed of 5 main modules:

- Moving modules: Robot and workstation.
- Static modules: The grid and bin.
- Command center: Controller.

• **Workstations:** They present our inputs and outputs of the system. The empty bins are replenished in those ports, and then transported to the correspondent position on our matrix. Another workstation from the destination side will allow the arrival of the bin to the hands of the operator whenever needed.

• **Bin:** The bins are made to hold the pieces in the inventory. They are stored next and on top of each others and their dimensions depends on the weight and dimensions of the parts.

• **Grid:** The grids are the base of our system, the pillars hold the bins and allow the movement of the robots by providing tracks to drive on.

• **Robot:** On the top of the grids, the robots will be driving and retrieving the bins, the choice of the correct bin will depend on the order given to the controller. The robots are wirelessly connected and charge themselves when they are not in use (in charging stations).

Application of auto store in MRA2 Project:

The biggest problem we had is about the transport of the big parts, especially the distribution housing semi assembled parts, from the correspondent assembly lines to the buffer area, crossing the street, and then after, to the backbone (final assembly line), added to that the huge storage space needed for the semi finished goods.

With this technology, there will be no more time wastes for replenishment of the trolleys and transporting them.

It all begins from the assembly line output, where the pre-assembled part is put by the operator in the empty bin waiting in the workstation (port), which is an ergonomically adjusted, safe, and easily controlled workstation. When the bin is full, it is lifted by the elevator connected to the robot and transported to the right coordination in the buffer area. Another empty bin will be ready in the workstation to lift the next part.

The bins are stored orderly based on the type of the part and the variety version, and also sorted by high and low runner criteria: each number of column will be allocated for a specific kind of material. Based on the customer demand, a daily production sequence of HVAC units is pre established so the operator and the buffer controller system know what they are and will be producing next, so the next bin is ready to be picked by the robot and delivered to the final assembly line (backbone).

As a result, no delays, no trolleys, no extra space, no change over time, no replenishment time and overburden will be generated, and so the material flow will be smooth in the assembly area. The storage area is expected to cover a surface of 18m x 9m = 162m² and the system will be installed for 3m Height usable of the warehouse, plus 2 free meters up for recharge of the robots and regular preventive maintenance.

This means, after discussion with the customer, it's estimated to have 4446 bins, and by comparing it with our maximum production daily, will be enough to handle the parts.

The figure 34 below shows how the system should look like and how the robots pick, store and transport the parts from and to the workstations. [17]



Figure 34. Cubic space creation: Autostore system.

Idea 2: Paternosters solution: Plan B:

´Paternoster´ is another option for pure storage activities. This solution is another use of air space for cubic storage, in order to exploit space resources by minimizing the existing storage space. The system is an elevator that consists of shelves moving in a loop with cycle chain up and down. The capacity of storage of this system will minimize our need for space. The paternoster system will insure a dynamic movement of the trays allowing, from one side, the replenishment of finished goods or different parts, and from another side, a feeding to the final assembly line and so a nonstop process for intralogistic.



Figure 35. Paternoster system

Both the ideas of auto store and paternoster solutions will bring new benefits and gain in term of space, time and organisation of the flow and the material handling. Those ideas are being elaborated and under discussion and currently, the negotiations are in advancement with the suppliers.

Conclusion :

This Chapter was dedicated to my collaboration and added value to the project. It included modifications on the layout to get the best possible plan of the future assembly lines. And then calculations were made to know how the project will be evolving, and so we can manage the growth of the demand and respond to the quantity asked by the customer and especially manage our space and the WIP. The chapter was concluded by bringing the best options to manage the information flow and explaining the process to install the E-Kanban process.

New ideas to manage our material and information flow were introduced, which brings us to the next point which is the possibilities of investment in those idea.

General conclusion and prospects:

Aiming to stand out in the local, as well as international market, Valeo has been adapting new and innovative conceptions of assembly lines, based mostly on the automatic operation, stepping more into the factory of the future strategy.

My internship within Valeo Climate Control was an opportunity to understand, to study and to carry out my knowledge in the equipments assembly field within the automotive industry.

In fact, my work allowed to deepen the theoretical knowledge in methods of production and to better understand the lean approach in automotive industry field.

This opportunity made the process of HVAC unit production easier to understand and so I had a clearer image about the differences between the varieties and the production process.

This study opened the way to deeper understanding of the details of assembly as well as the to the criticality of each step, and how the management of parts needs more focusing on the movement of the physical material, in order to avoid problems such as delays and non quality and ensure an effective reaction to the changes.

The first part of this work was devoted to a presentation of the host company Valeo, and its strategies, targets and vision on the short and long term as an automotive supplier.

Then, the second part was dedicated to the functionalities of the climate control systems, their utilities, categories and the main parts composing the final HVAC. This has led particularly to the description of our project entitled MRA2 and the different points to focus on. Those points describe the framework of my work which articulates around the new assembly lines implementation, and eventually the intra-logistics and the intern material and information flow between the assembly lines.

The mentionned points were detailed in the last part of this work, in which I presented my intervention to bring modifications on the existing temporary layout and then just after, new solutions were brought to manage the material flow, using existing tools and means, and also other ideas that need new investments for more optimized and effective results.

The logical outcome of this research would be to extend this initiative to identify the adapted level of use of the correspondent tools in each step.

Other than the fact that we were able to create a complete production plan for a new and highly innovative project within the assembly area, the main achievement realised during this internship is the fact that we were able to manage the implementation of new assembly lines

with a potential smooth and organised material and information flow planning, by a studied combination of existing materials, means and exploited potentials already in the company (means of transportation, loading, softwares, information systems, tools…), which adds more value to our project with less costs (low investments).

Finally, it's important to mention that this case study about the production of HVAC units and the implementation of the parts and the tasks related to this issue, represents the first link in the chain of improvement that will be followed by the installation of the assembly lines and then a testing period of the efficiency of the lines, and this will allow a detection of failures or the further improvement to bring to the lines.

Bibliography and references list:

Web sites :

[1] https://www.valeo.com (conselted on 18/08/2018)

[2] https://www.valeo.com/key-figures/ (conselted on 18/08/2018)

[3] https://www.valeo.com/our-activities/ (conselted on 18/08/2018)

[4] http://www.valeo.de/en/valeo-in-germany/valeo-activities-in-germany.html (conselted on 18/08/2018)

[5] http://www.Smart workplace.apps.valeo.com/home/ths-in-press-tomorrow climate control (conselted on 19/08/2018)

[6] https://qualite.ooreka.fr/comprendre/5s (conselted on 22/07/2018)

[7] http://leleanmanufacturing.com/les-7-gaspillages/ (conselted on 23/07/2018)

[8]http://leanmanufacturingtools.org/551/creating-a-value-stream-map/ (conselted on 06/07/2018)

[9]http://www.leanaccountants.com/2011/12/what-is-smed-and-why-is-it-important.html (conselted on 12/06/2018)

[10]http://asq.org/learn-about-quality/process-analysis-tools/overview/flowchart.html (conselted on 16/07/2018)

[11] http://www.six-sigma-material.com/Spaghetti-Diagram.html (conselted on 05/06/2018)

[12] https://www.isixsigma.com/methodology/kaizen/kaizen-six-sigma-ensures-continuous-improvement/ (conselted on 30/06/2018)

[13] https://kanbanize.com/continuous-flow/heijunka/ (conselted on 30/08/2018)

[14] https://www.graphicproducts.com/articles/pull-system/ (conselted on 08/08/2018)

[15]https://www.toyotaglobal.com/company/vision_philosophy/toyota_production_system/just-in-time.html (conselted on 16/06/2018)

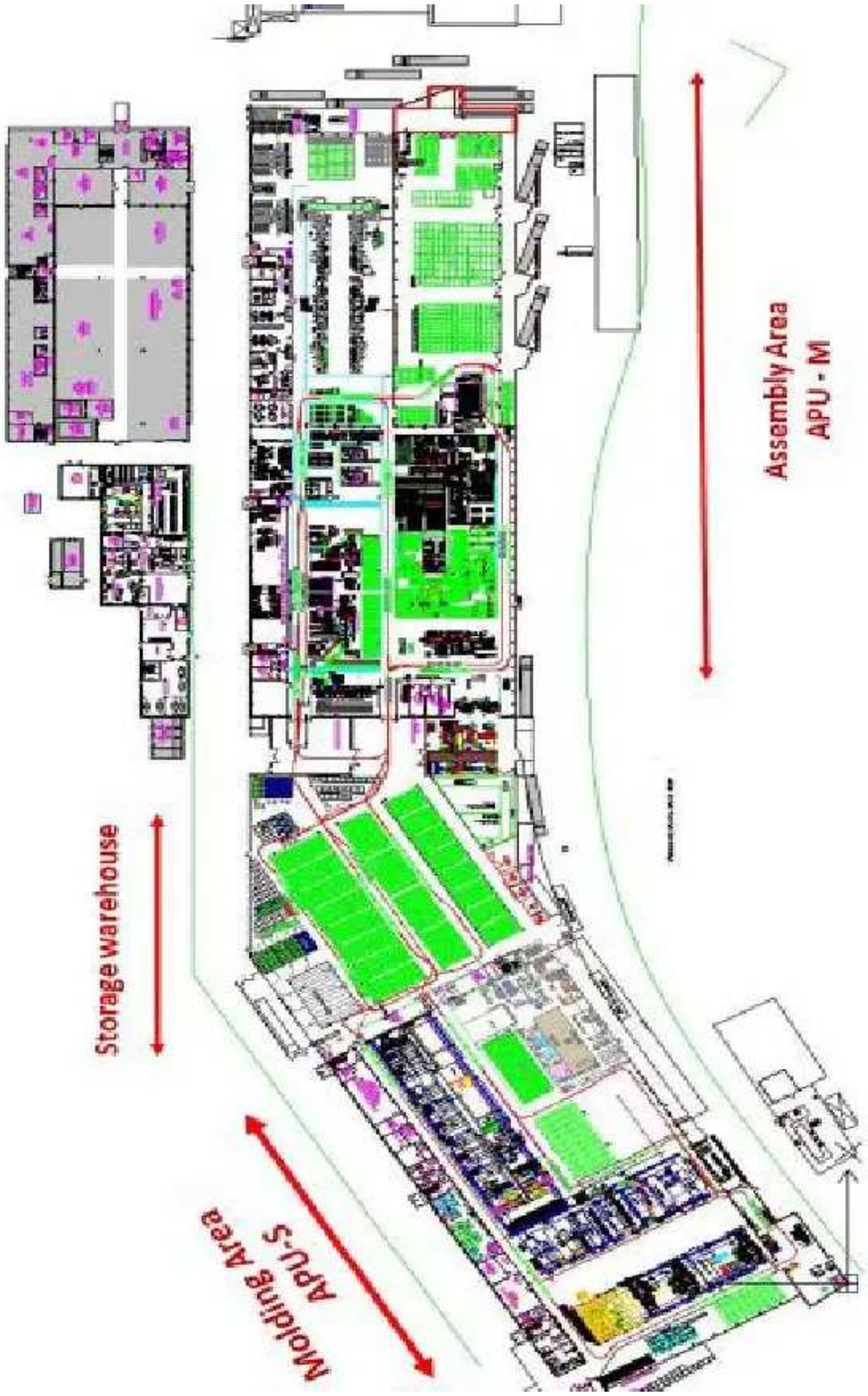
[16] www.accountingtools.com (conselted on 23/08/2018)

[17] https://www.autostoresystem.com/ (conselted on 03/09/2018)

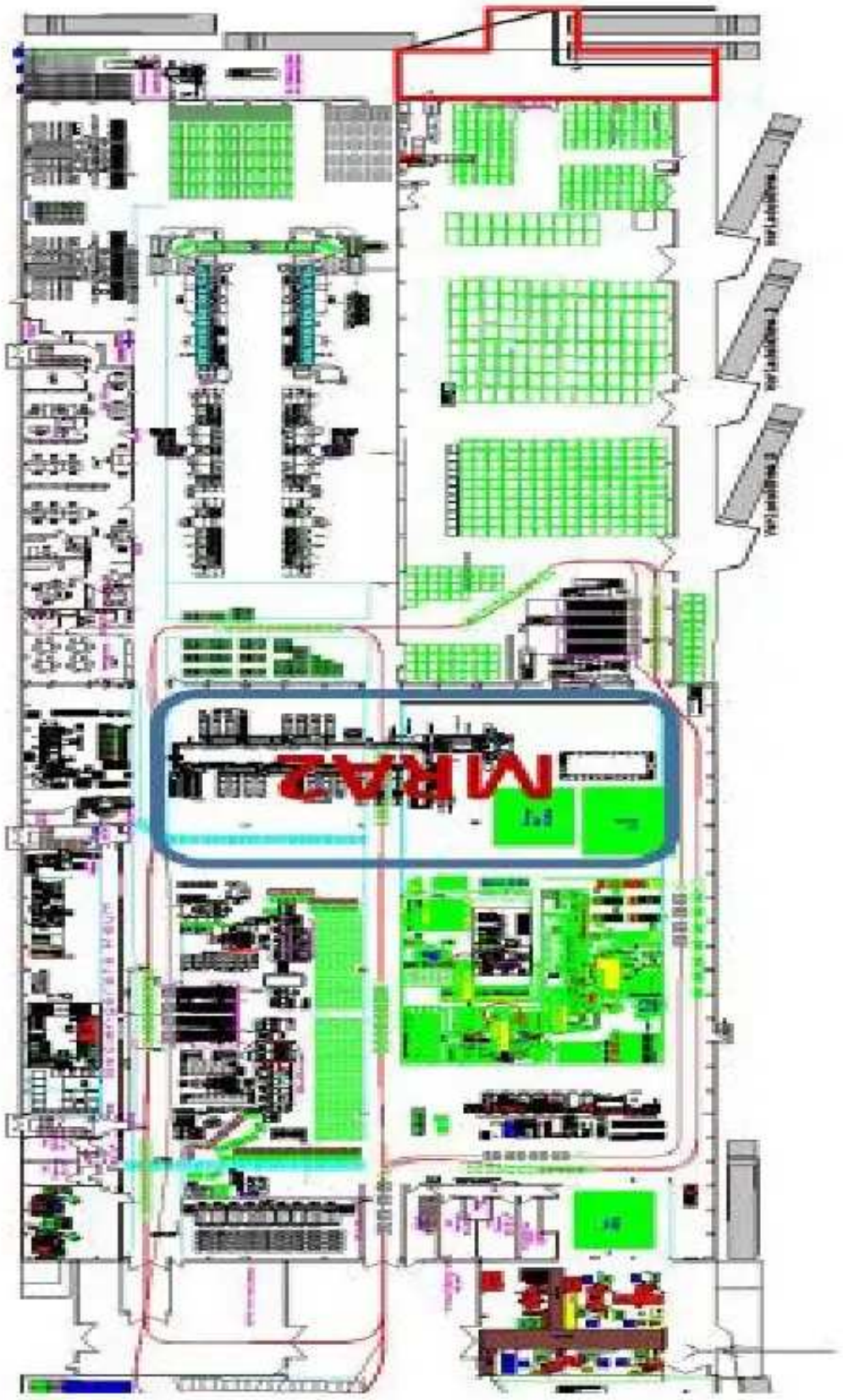
Case studies and documents :

- Eric Karlsson – Tobias Thoresson (2011): A comparative study of the material feeding principles kitting and sequencing at SAAB Automobile, Trollhättan: creation of guiding principles of which articles to be supplied with kitting.
- Babara Lyonnet (2010): Amelioration de la performance industrielle: vers un systeme de production lean adapte aux enterprises du pole de competitivite.
- Grant Mackerron, Maneesh Kumar, Vikas Kumar, Ann Esain: Rplenishment policy using e-kanban: a framework for successful implementation.
- Meriem Lafou – 11 July 2016 : Contribution a la conception de systeme d'assemblage automobile, performants, perennes et innovants par des indicateurs technologiques et economiques repondant a la diversite croissante des produits.
- Anis Said – 20 June 2013 : Amelioration de la productivite d'une ligne d'assemblage de la nouvelle Sandero et Maitrise de l'indicateur de respect de l'ordre d'enchainement des vehicules.
- Elin Anderson - Louise Holmgren – 2015 : Improving Efficiency of the material supply in assembly lines – a case study in the automotive industry in southern.
- Melany Reyneke – 2010 : Assembly line layout and process optimization at Powertech Transformers.

Appendixes :



Appendix 1. Valeo Production departments.



Appendix 2. Assembly Area : APU-M

Station 1 :	Kinematic insertion and actuator pre-fixation.
Station 2 :	Small parts (levers, linkages, shafts) pre-fixation.
Station 3 :	Flap pre-fixation.
Station 4 :	Housing outer 1.
Station 5 :	Flap assembly 1.
Station 6 :	Housing inner.
Station 7 :	Flap assembly 2 (Pre-assembly mixing).
Station 8 :	Housing centerwall.
Station 9 :	Flap assembly 3 (Pre-assembly mixing).
Station 10 :	Housing inner 2.
Station 11 :	Flap assembly 4.
Station 12 :	Housing outer 2.
Station 13 :	Housing finishing TAHC side.
Station 14 :	Actuator assembly and screwing.
Station 15 :	180° - part rotation.
Station 16:	Housing finishing kinematic side.
Station 17 :	Wiring and test station.

Appendix 3. Distribution Housing: Process functions.

Section 1 :	Step 1 :	Insert Housing.
	Step 2 :	Automatic assembly of greased fresh air flap.
	Step 3 :	Automatic assembly of greased recirculation flap.
	Step 4 :	Automatic assembly of levers (x2)
	Step 5 :	Manual assembly of fresh air seal.
Section 2 :	Step 6 :	Manual Pre-fix kinematic.
	Step 7 :	Automatic assembly of actuator to sub-assembly. Automatic screwing of kinematic (x3) and actuator (2x).

Air Intake MRA2: Process functions.

Section 1 :	Step 1 :	Housing insertion and greasing.
	Step 2 :	Assembly of 3 flaps.
	Step 3 :	Clip housing right to housing left.
	Step 4 :	Assembly of flap levers to flaps (x3)
	Step 5 :	Assembly of fresh air seal to sub assembly.
Section 2 :	Step 6 :	Assembly of air intake kinematic to sub-assembly and link coupling rods to flap levers.
	Step 7 :	Automatic assembly of actuator to sub-assembly. Automatic screwing of kinematic (x3) and actuator (2x).

Air Intake EVA 2: Process functions.

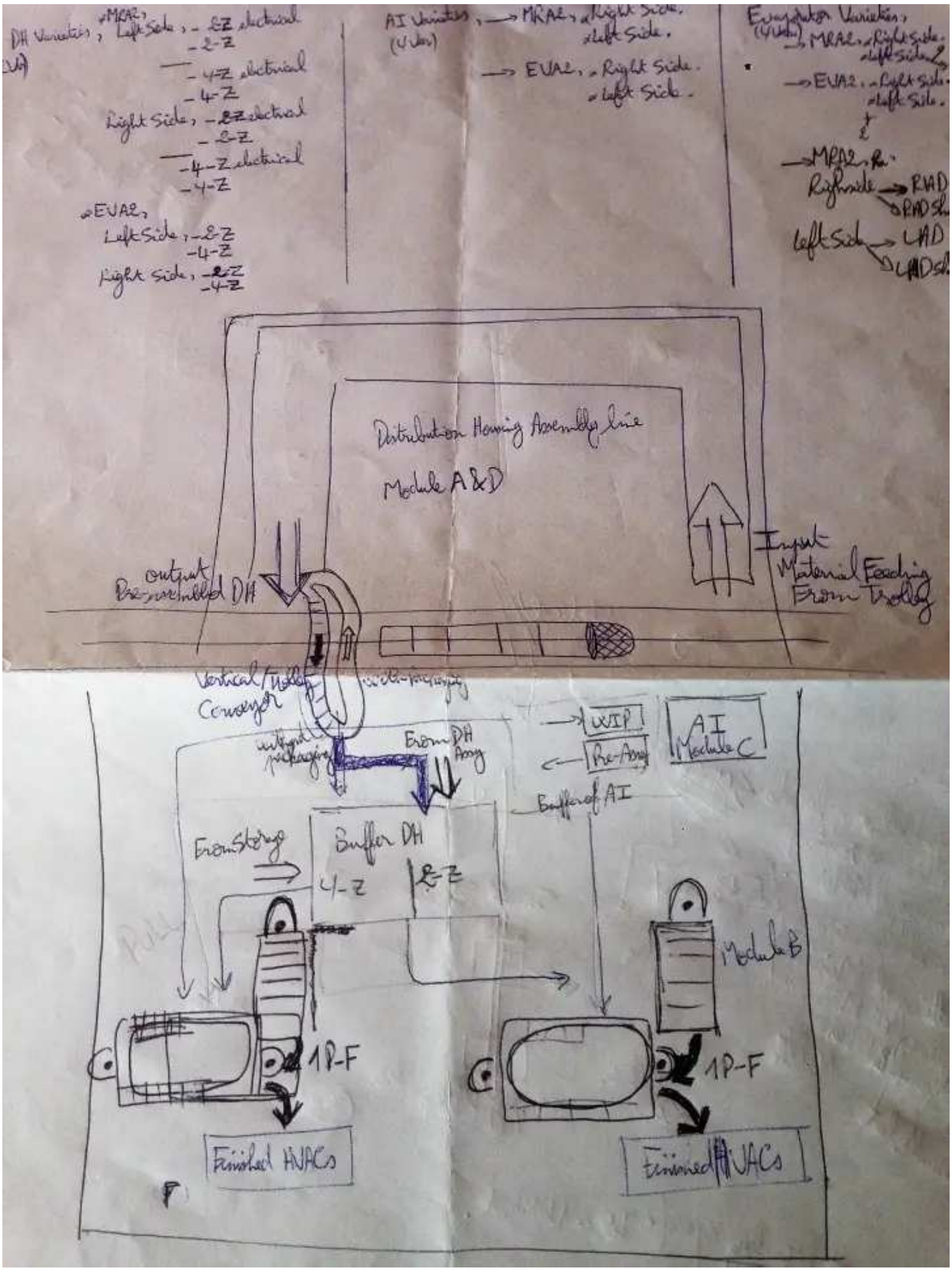
Appendix 4. Air Intake process function for MRA2 and EVA2.

Workstation 1:	Manual assembly station: Picking and placing of housings into the supports of the device as well as assembly lines.
Workstation 2:	MRA Clipping station: Housing clipping of BR 206 LHD/RHD variants.
Workstation 3:	EVA 2 LHD/RHD clipping station: Housing clipping of EVA 2 LHD/RHD variants.
Workstation 4:	Automatic screwing station.
Workstation 5:	Buffer places and picking station.

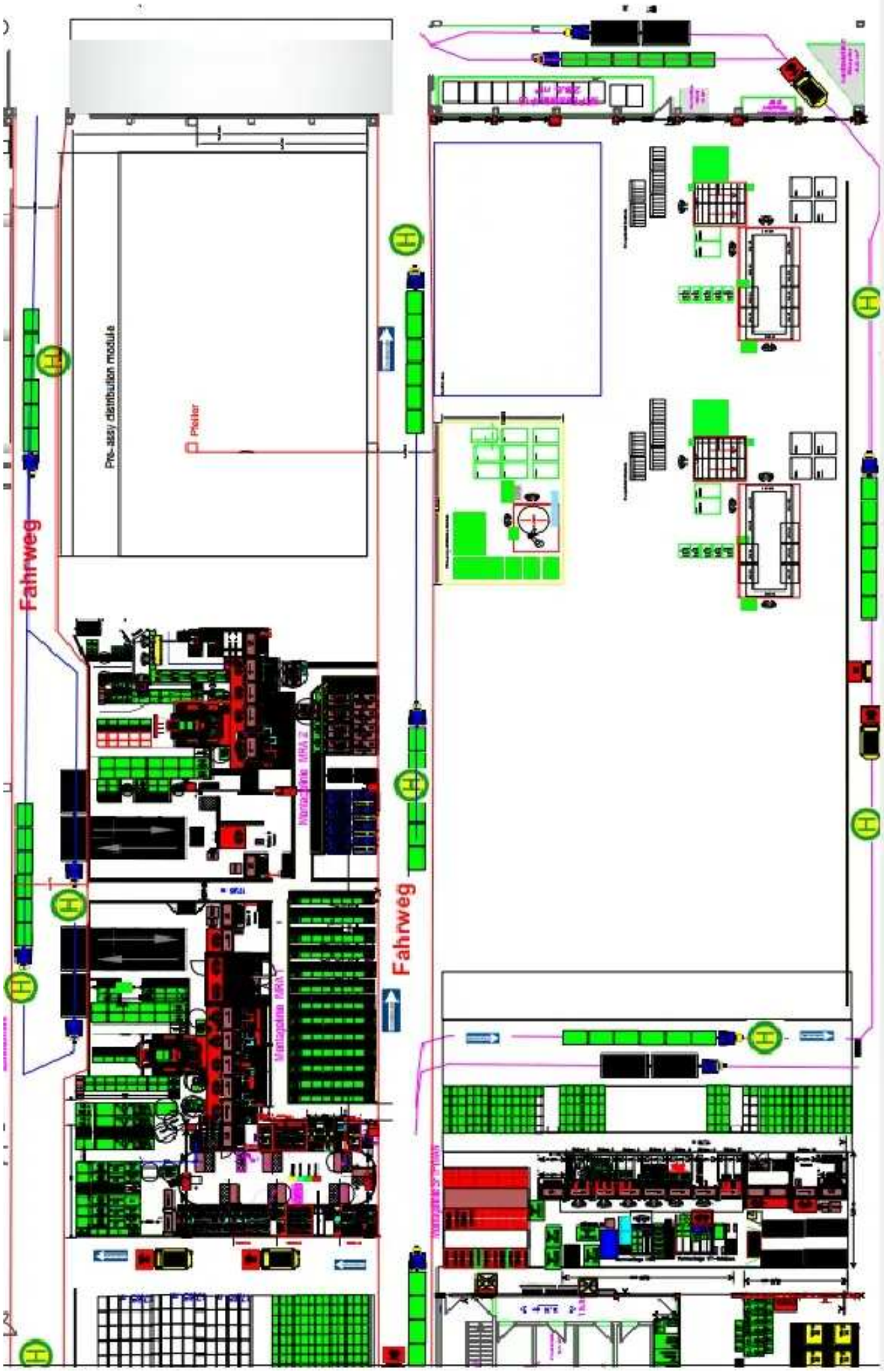
Appendix 5. Evaporator: Process functions.

Workstation 1:	Evaporator Module Loading and HVAC Packaging Station (Line Start / End)
Workstation 2:	Distribution Module Assembly Station (Wedding Station)
Workstation 3:	Air intake Module Assembly Station
Workstation 4:	Automatic Screwing Station
Workstation 5:	Automatic Leak Test Station
Workstation 6:	Manual Assembly Station
Workstation 7:	1st EOLT Test Station
Workstation 8:	2nd EOLT Test Station w/ Noise Chamber
Workstation 9:	3rd EOLT Test Station

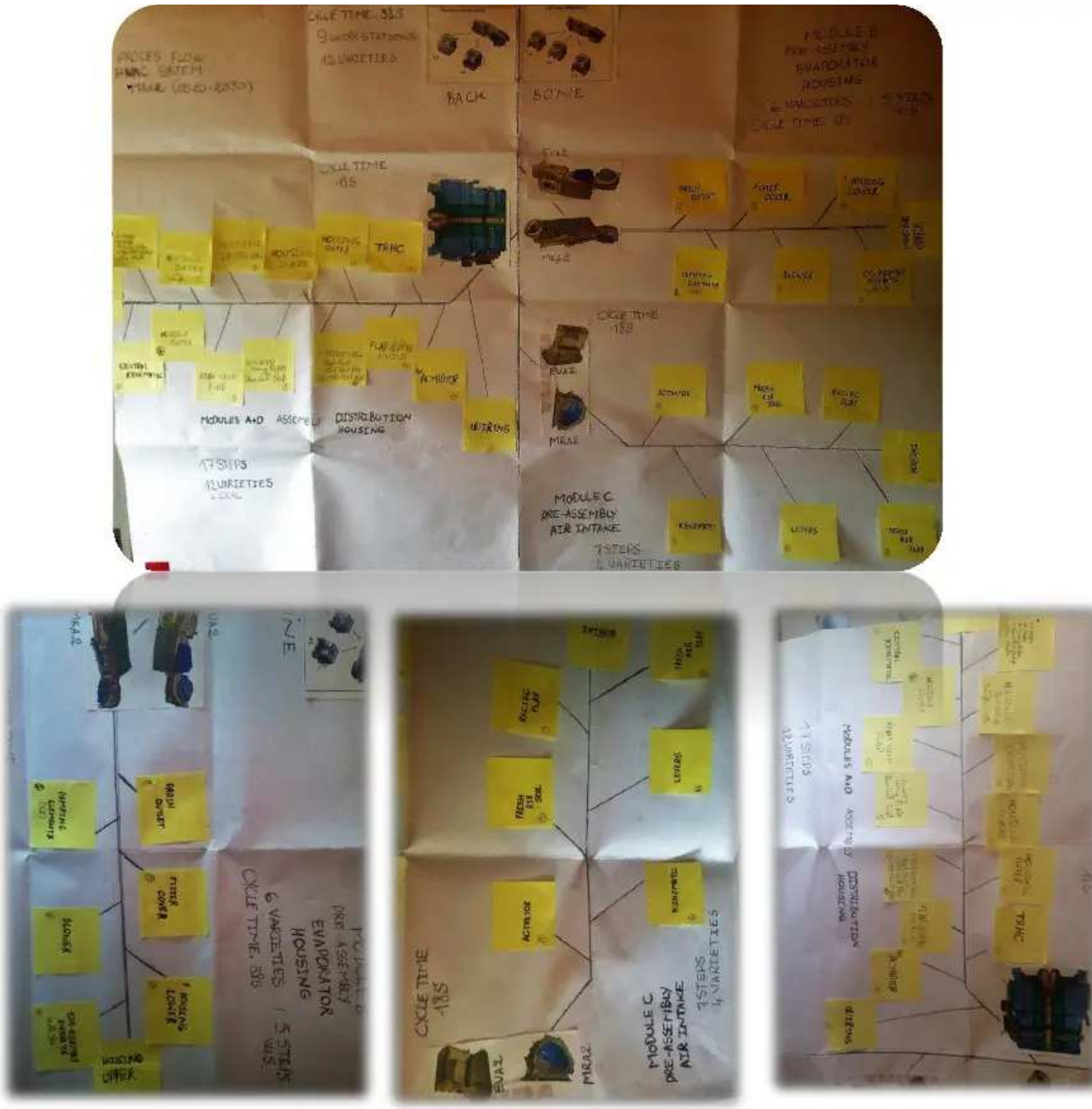
Appendix 6. Backbone: Process functions.



Appendix 7: Layout and assmebly lines handly elaboration.



Appendix 8. New MRA2 Layout.



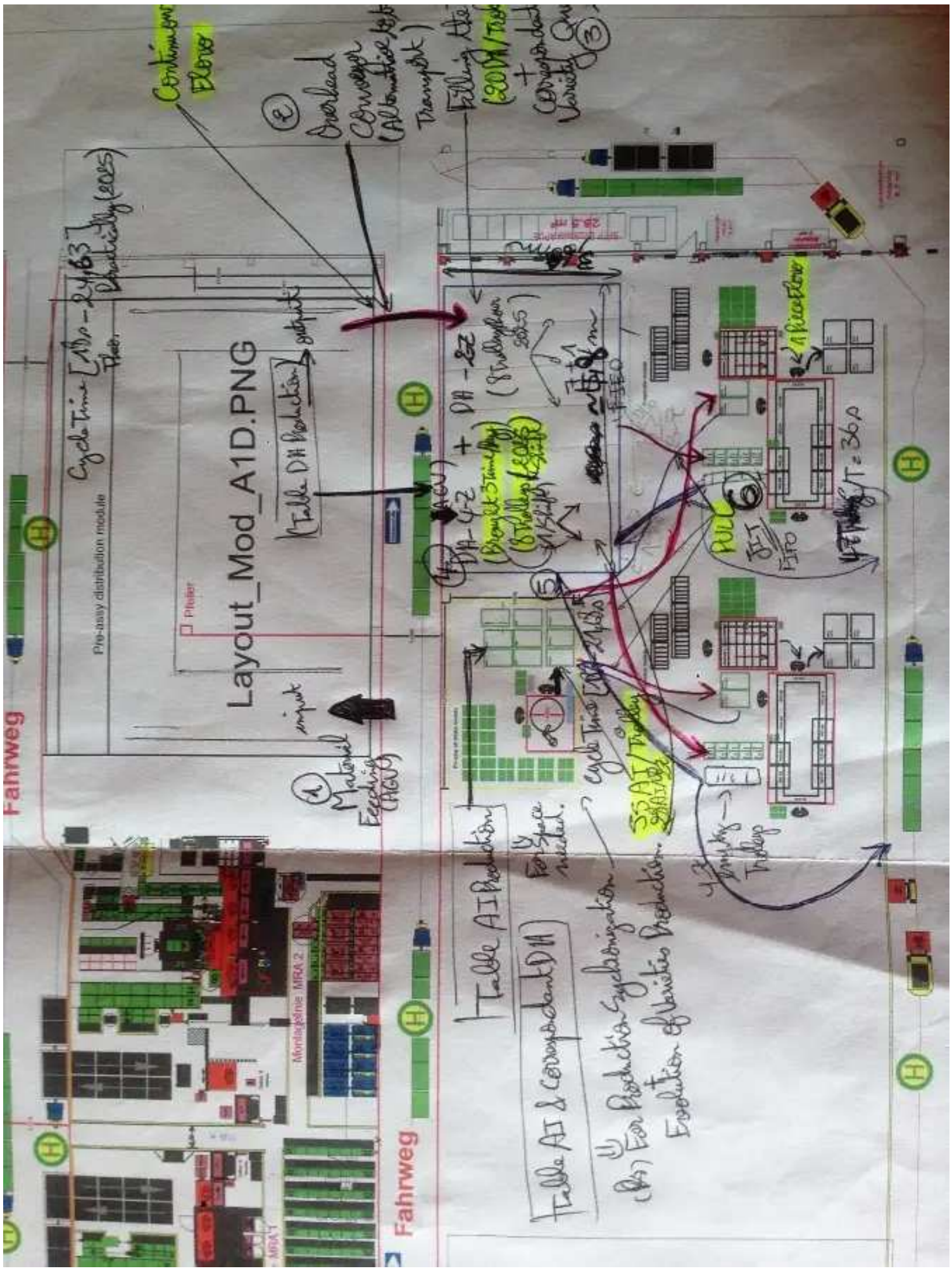
Appendix 9. Ishikawa Diagram for assembly process.

Appendix 10. DH production and space calculation for buffer area.

Year	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Total HVAC Production	1 346	225 877	451 750	817 057	899 667	910 124	853 186	751 581	561 490	265 184	94 026
Production/Day	5	892,794466	1799,8008	3281,35341	3584,33068	3640,496	3385,65873	2970,67589	2254,97992	1060,736	376,104
Production/Shift		297,598155	599,933599	1093,78447	1194,77689	1213,49867	1128,55291	990,225296	751,659973	353,578667	125,368
Production/Hour		40,7668706	82,1826848	149,833489	163,668067	166,232694	154,596289	135,647301	102,96712	48,4354338	17,1736986
Round		41	83	150	164	167	155	136	103	49	18
DH Production Bad Rodach	1 274	211 750	415 010	747 398	803 264	800 242	750 180	658 319	485 682	221 520	69 310
Production/Day	5	836,980237	1653,42629	3002,80321	3200,26295	3200,968	2976,90476	2602,01581	1948,12048	886,08	277,24
Production/Shift		278,993412	551,142098	1000,9344	1066,75432	1066,98933	992,301587	867,338603	649,373494	295,36	92,4133333
Production/Hour		38,2182757	75,4989176	137,114302	146,130728	146,162922	135,931724	118,813507	88,9552731	40,460274	12,6593607
Round		39	76	138	147	147	136	119	89	41	13
Storage											
Number of Trolleys/Day	1	41,8490119	82,6713147	150,140161	160,013147	160,0484	148,845238	130,100791	97,4060241	44,304	13,862
Round	1	42	83	151	161	161	149	132	98	45	14
Space/Day (1 Trolley= 0,54 m²) (m²)	1	22,68	44,82	81,54	86,94	86,94	80,46	71,28	52,92	24,3	7,56
Number of Trolleys/Shift		13,9496706	27,5571049	50,0467202	53,3377158	53,3494667	49,6150794	43,3669302	32,4686747	14,768	4,62066667
Round		14	28	51	54	54	50	44	33	15	5
space/Shift (m²)		7,56	15,12	27,54	29,16	29,16	27	23,76	17,82	8,1	2,7
Number of Trolleys/Hour		1,91091378	3,77494588	6,8557151	7,30653641	7,30814612	6,79658621	5,94067537	4,44776366	2,0230137	0,63296804
Round		2	4	7	8	8	7	6	5	3	1
space/Hour (m²)		1,08	2,16	3,78	4,32	4,32	3,78	3,24	2,7	1,62	0,54
DH scored from Rannovik	72	14 122	36 740	69 359	96 403	109 883	103 006	93 273	76 407	43 665	24 716
Storage/Day	0	55,8181818	146,374502	278,550201	384,067729	439,532	408,753968	368,660079	306,855422	174,66	98,864
Storage/Shift		18,6060606	48,7915007	92,8500669	128,022576	146,510667	136,251323	122,886693	102,285141	58,22	32,9546667
Storage/Hour		2,54877543	6,68376721	12,7191873	17,5373392	20,0699543	18,6645648	16,8337936	14,0116631	7,97534247	4,5143379
Storage											
Number of Trolleys/Day		2,79090909	7,3187251	13,92751	19,2033865	21,9766	20,4376984	18,433004	15,3427711	8,733	4,9432
Round	1	3	8	14	20	22	21	19	16	9	5
Space/Day (m²)	1	1,62	4,32	7,56	10,8	11,88	11,34	10,26	8,64	4,86	2,7
Number of Trolleys/Shift		0,93030303	2,43957503	4,64250335	6,40112882	7,32553333	6,81256614	6,14433465	5,11425703	2,911	1,64773333
Round		1	3	5	7	8	7	7	6	3	2
space/Shift (m²)		0,54	1,62	2,7	3,78	4,32	3,78	3,78	3,24	1,62	1,08
Number of Trolleys/Hour		0,12743877	0,33418836	0,63595936	0,87686696	1,00349772	0,93322824	0,84168968	0,70058315	0,39876712	0,22571689
Round		1	1	1	1	1	1	1	1	1	1
space/Hour (m²)		0,54	0,54	0,54	0,54	0,54	0,54	0,54	0,54	0,54	0,54
Total space/Shift (m²)		8,1	16,74	30,24	32,94	33,48	30,78	27,54	21,06	9,72	3,78
Total Space/hour (m²)		1,62	2,7	4,32	4,86	4,86	4,32	3,78	3,24	2,16	1,08

Appendix 11. Table of AI Production and space for the buffer area.

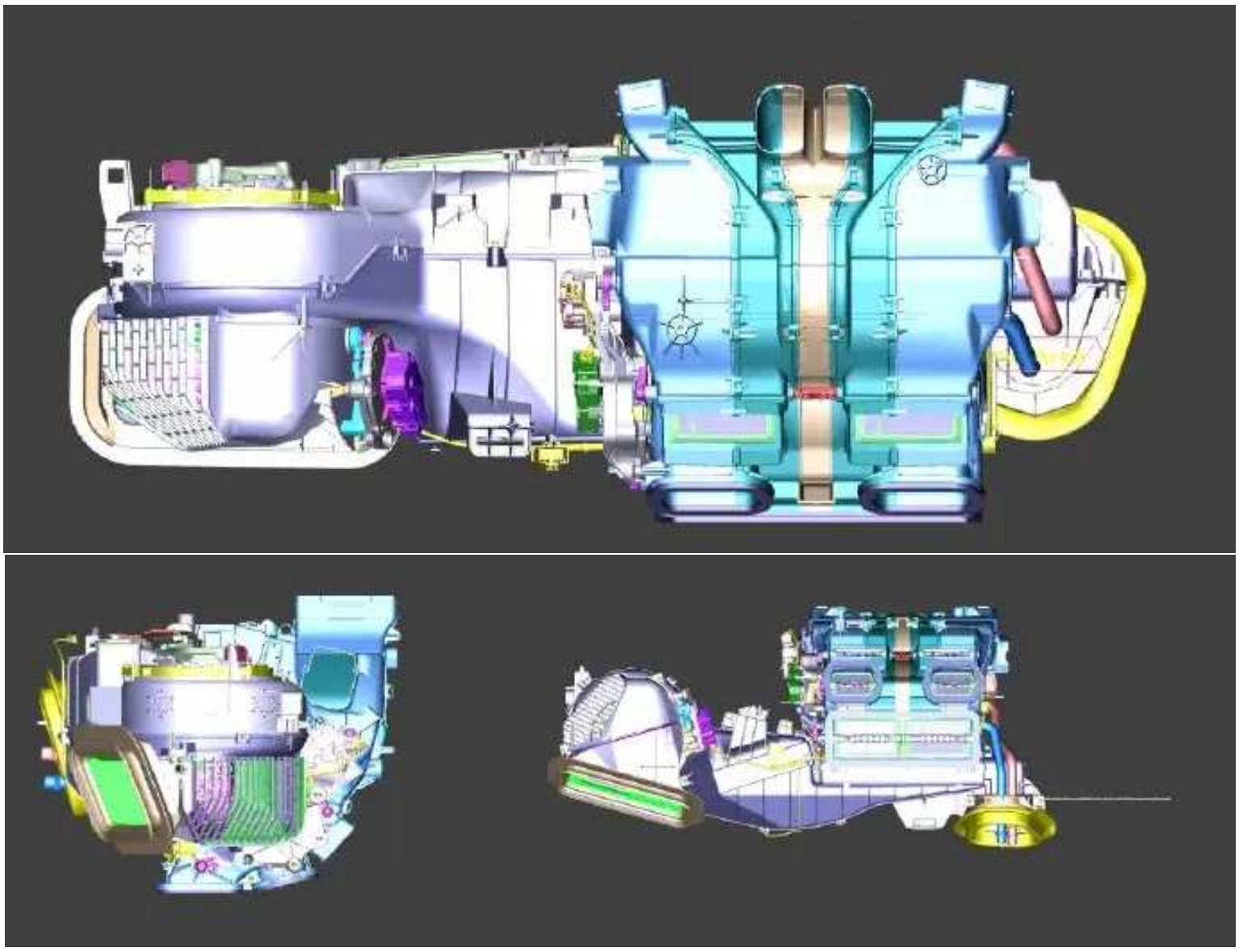
Year	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Total HVAC Production	1 346	225 877	451 750	817 057	899 667	910 124	853 186	751 581	561 490	265 184	94 026
Production/Day	5	892,794466	1799,8008	3281,35341	3584,33068	3640,496	3385,65873	2970,67589	2254,97992	1060,736	376,104
Production/Shift		297,598155	599,933599	1093,78447	1194,77689	1213,49867	1128,55291	990,225296	751,659973	353,578667	125,368
Production/Hour		40,7668706	82,1826848	149,833489	163,668067	166,232694	154,596289	135,647301	102,96712	48,4354338	17,1736986
Round		41	83	150	164	167	155	136	103	49	18
AI Production	1 346	225 877	451 750	817 057	899 667	910 124	853 186	751 581	561 490	265 184	94 026
Production/Day	5	892,794466	1799,8008	3281,35341	3584,33068	3640,496	3385,65873	2970,67589	2254,97992	1060,736	376,104
Production/Shift		297,598155	599,933599	1093,78447	1194,77689	1213,49867	1128,55291	990,225296	751,659973	353,578667	125,368
Production/Hour		40,7668706	82,1826848	149,833489	163,668067	166,232694	154,596289	135,647301	102,96712	48,4354338	17,1736986
Round		41	83	150	164	167	155	136	103	49	18
Storage											
Number of Trolleys/Day	1	31,8855167	64,2785999	117,191193	128,01181	130,017714	120,916383	106,095567	80,5349971	37,8834286	13,4322857
Round	1	32	65	118	129	131	121	107	81	38	14
Space/Day (1Trolley= 0,528 m^2) (m^2)	0,528	16,896	34,32	62,304	68,112	69,168	63,888	56,496	42,768	20,064	7,392
Number of Trolleys/Shift		10,6285056	21,4262	39,0637311	42,6706033	43,3392381	40,3054611	35,3651892	26,844999	12,6278095	4,47742857
Round		11	22	40	43	44	41	36	27	13	5
space/Shift (m^2)		5,808	11,616	21,12	22,704	23,232	21,648	19,008	14,256	6,864	2,64
Number of Trolleys/Hour		1,45595966	2,93509589	5,35119604	5,84528812	5,93688193	5,52129604	4,84454646	3,67739713	1,72983692	0,61334638
Round		2	3	6	6	6	6	5	4	2	1
space/Hour (m^2)		1,056	1,584	3,168	3,168	3,168	3,168	2,64	2,112	1,056	0,528



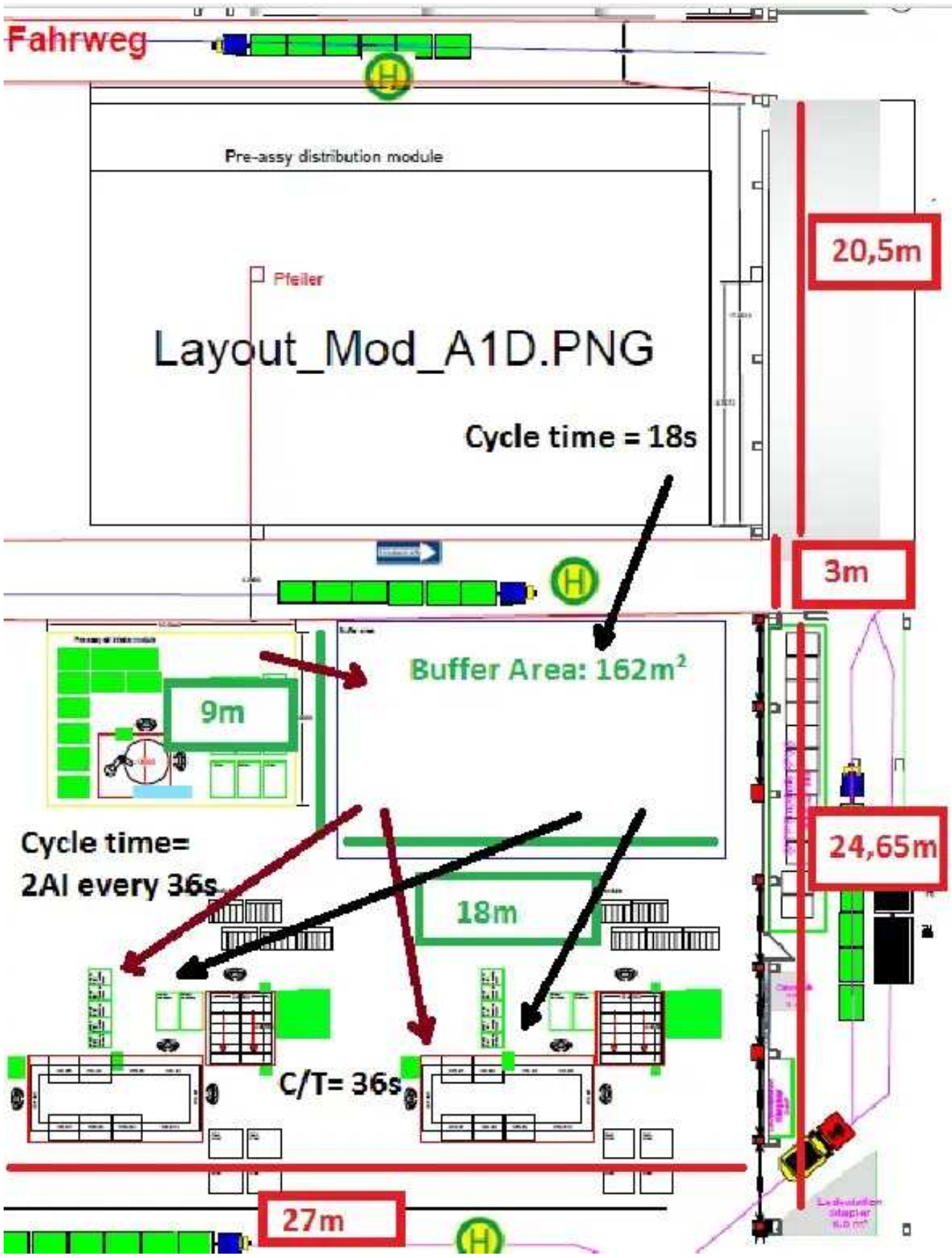
Appendix 12. Spaghetti Diagram.

Appendix 13. Air Intake and correspondent Distribution Housing Varieties.

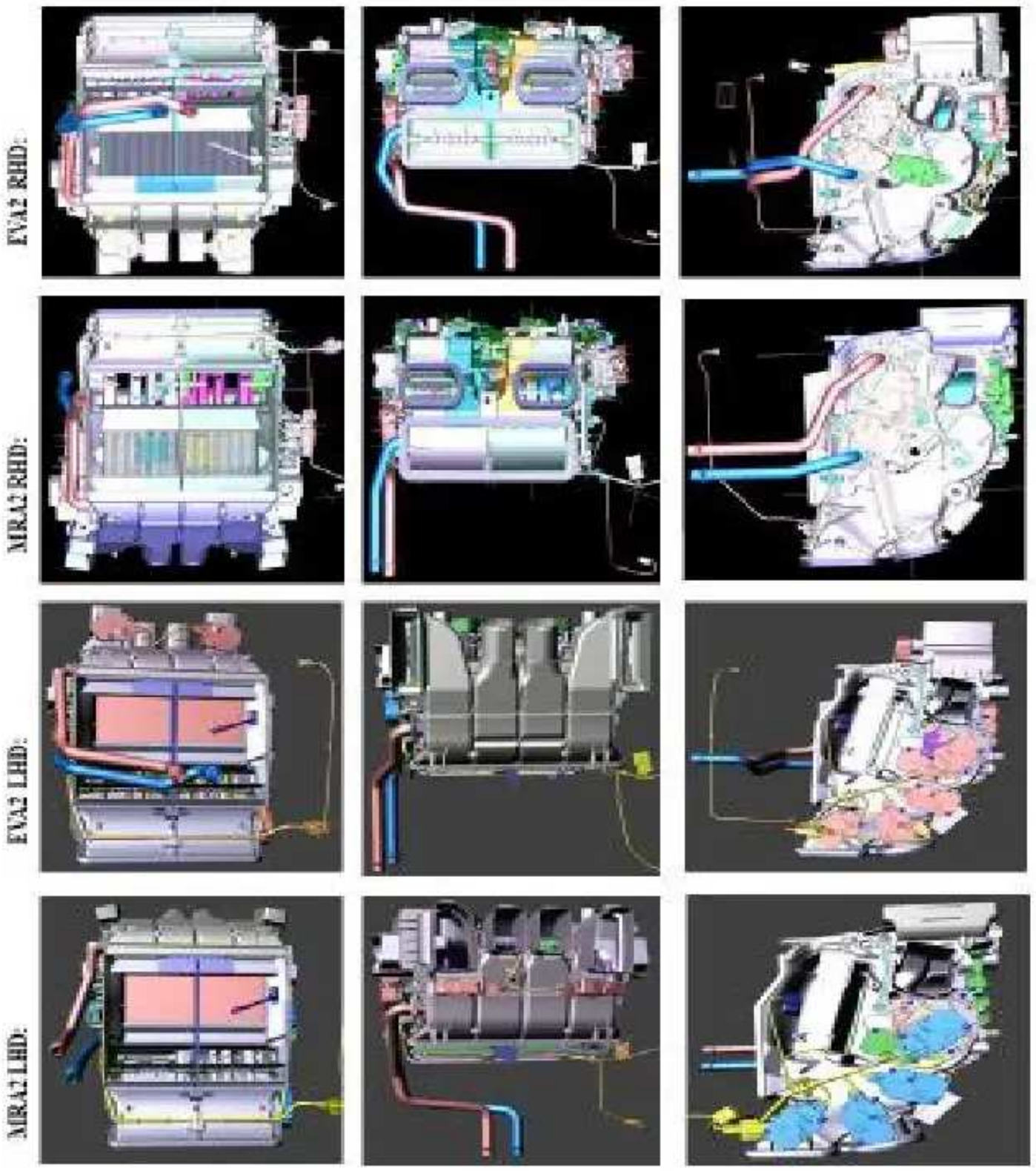
Air Intake	D. Housing	2020	2020	2021	2021	2022	2022	2023	2023	2024	2024	2025	2025	2026	2026	2027	2027	2028	2028	2029	2029	2030	2030
MRA2 LHD	MRA2 LHD 2 Zones	926	875	153 132	143646	284 941	264762	541 159	500125	591 416	531363	603 490	532683	562 997	496997	494 711	435026	380 485	329576	196 683	164 406	75 136	55 631
	MRA2 LHD 4 Zones		52		9 485		20 180		41 033		60 052		70 807		66 001		59 685		50 909		32 275		19 505
MRA2 RHD	MRA2 RHD 2 Zones	370	357	56656	54433	91865	86547	157961	147327	173158	157072	173337	154257	162255	144440	140919	124725	104688	90637	52468	43483	18890	13 678
	MRA2 RHD 4 Zones		13		2 222		5 319		10 635		16 086		19 081		17 815		16 194		14 050		8 984		5 211
EVA2 LHD	EVA2 LHD 2 Zones	44	37	13997	11 898	65201	55 421	102605	87 214	117532	99 902	115968	98 573	111303	94 607	100877	85 746	66395	56 436	13950	11 858	0	0
	EVA2 LHD 4 Zones		7		2 100		9 780		15 391		17 630		17 395		16 695		15 132		9 959		2 093		0
EVA2 RHD	EVA2 RHD 2 Zones	7	6	2092	1 778	9743	8 281	15332	13 032	17562	14 928	17329	14 729	16631	14 137	15074	12 813	9921	8 433	2085	1 772	0	0
	EVA2 RHD 4 Zones		1		314		1 461		2 300		2 634		2 599		2 495		2 161		1 488		313		0
Daily production																							
	Working Days	253		253		251		249		251		250		252		253		249		250		250	
Air Intake	D. Housing	2020	2020	2021	2021	2022	2022	2023	2023	2024	2024	2025	2025	2026	2026	2027	2027	2028	2028	2029	2029	2030	2030
MRA2 LHD	ZB Klimakasten LL 2Z	4	3	605	567,77	1 135	1054,8	2 173	2008,5	2 356	2117	2 414	2130,7	2 234	1972,2	1 955	1719,5	1 528	1323,6	787	658	301	223
	ZB Klimakasten LL 4Zonen		0		37		80		165		239		283		262		236		204		129		78
MRA2 RHD	ZB Klimakasten RL 2Z	1,4625	1,4111	223,94	215,15	366	344,81	634,38	591,67	689,87	625,78	693,35	617,03	643,87	573,17	556,99	492,98	420,43	364	209,87	173,93	75,56	55
	ZB Klimakasten RL 4Zonen		0		9		21		43		64		76		71		64		56		36		21
EVA2 LHD	ZB Klimakasten LL 2-	0,1739	0	55,324	47	259,76	221	412,07	350	468,25	398	463,87	394	441,68	375	398,72	339	266,65	227	55,8	47	0	0
	ZB Klimakasten LL 4-Zonen		0		8		39		62		70		70		66		60		40		8		0
EVA2 RHD	ZB Klimakasten RL 2-	0	0	8,2688	7	38,817	33	61,574	52	69,968	59	69,316	59	65,996	56	59,581	51	39,843	34	8,34	7	0	0
	ZB Klimakasten RL 4-Zonen		0		1		6		9		10		10		10		9		6		1		0



Appendix 14. HVAC Units in 3D.



Appendix 15. Assembly flow.



Appendix 16. Distribution Housing Varieties.

العنوان: إدارة تدفق المواد لموقع الإنتاج.

ملخّص:

يندرج هذا العمل في إطار مشروع ختم الدروس للحصول على شهادة الهندسة من المعهد الوطني للعلوم التطبيقية والتكنولوجيا.
تم إنجاز هذا المشروع بشركة "فالبو" بألمانيا و يدور المشروع حول إنتاج وتجميع أنظمة تكييف هواء السيارات و يهدف بالأساس إلى بناء وضبط خطوط التجميع ، بالإضافة إلى دراسة وتخطيط تدفق المواد والتدفق المقابل للمعلومات.
يوضح هذا التقرير الخطوات الضرورية للتنفيذ بالإضافة إلى الحلول المتكاملة لتحسين تدفق المواد والمعلومات.

المفاتيح :

خطوط التجميع / تدفقات المواد والمعلومات / اللوجستية الداخلية

Titre : Gestion des flux de matières pour le site de production.

Résumé :

Le présent travail, effectué chez Valeo: systèmes de climatisation à Bad Rodach, Allemagne, s'inscrit dans le cadre du projet de fin d'études pour l'obtention du Diplôme National d'Ingénieur en Instrumentation et Maintenance Industrielle à l'Institut National des Sciences appliquées et de la Technologie.

Le projet réalisé s'articule autour de la production et l'assemblage des systèmes de climatisation automobile pour le client « Daimler », et dont l'objectif se résume à la réalisation et l'adaptation des lignes d'assemblages, ainsi qu'une étude et planification du flux de matière et du flux d'information correspondant. Ce rapport détaille les étapes nécessaires pour l'implémentation ainsi que les solutions intégrées pour optimiser le flux de matière et d'information.

Mots-clés:

Lignes d'assemblage / Flux de matière et d'information / Intra-logistic / Produits en cours / Lean management

Title: Material Flow Management for the Production site.

Abstract:

This graduation project was realized within Valeo: air conditioning systems in Bad Rodach site in Germany, in order to obtain the National Diploma of Engineering in Instrumentation and Industrial Maintenance.

The project is based on the production and assembly of automotive air conditioning systems for the new "Daimler" vehicles, and is a solution for the adaptation of assembly lines, as well as a study and planning of the flow of material and the corresponding information flow.

This report details the steps followed for the implementation and brings integrated solutions to optimize the material and information flow.

Key Word:

Assembly lines / Material and Information flow / Intra-logistic / Work in Process / Lean management.

Réalisé à :

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